United States Patent Application Publication 20250287582
Kind Code A1
Publication Date September 11, 2025
Inventor(s) Jeong; Moonyoung et al.

SEMICONDUCTOR DEVICE

Abstract

A semiconductor device includes a first structure including a memory region and a first peripheral circuit region; and a second structure overlapping the first structure vertically and including a core circuit region and a second peripheral circuit region, wherein the memory region includes a first cell vertical active pattern; a first word line having a side surface facing a first side surface of the first cell vertical active pattern; and a first cell gate dielectric layer between the first cell vertical active pattern and the first word line, wherein the first peripheral circuit region includes a first peripheral vertical active pattern disposed at substantially the same level as the first cell vertical active pattern; a first peripheral gate electrode having a side surface facing a first side surface of the first peripheral vertical active pattern; and a first peripheral gate dielectric layer between the first peripheral vertical active pattern and the first peripheral gate electrode.

Inventors: **Jeong; Moonyoung (Suwon-si, KR), Lee; Sangho (Suwon-si, KR), Han; Joon (Suwon-si, KR)**

Applicant: **Samsung Electronics Co., Ltd.** (Suwon-si, KR)

Family ID: **1000008333178**

Assignee: **Samsung Electronics Co., Ltd. (Suwon-si, KR)**

Appl. No.: **18/965980**

Filed: **December 02, 2024**

Foreign Application Priority Data

KR	10-2024-0033051	Mar. 08, 2024

Publication Classification

Int. Cl.: **H10B12/00** (20230101)

U.S. Cl.:

CPC **H10B12/50** (20230201); **H10B12/315** (20230201); **H10B12/482** (20230201); **H10B12/488** (20230201);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims benefit of priority under 35 U.S.C. § 119(a) to Korean Patent Application No. 10-2024-0033051 filed on Mar. 8, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

FIELD

[0002] Example embodiments of the present disclosure relate to a semiconductor device and a method of manufacturing the same.

BACKGROUND

[0003] Research has been conducted to reduce the sizes of elements included in a semiconductor device and to improve performance thereof. For example, in a DRAM, research has been conducted to reliably and stably form elements having reduced sizes, but as sizes of the elements are reduced, performance of the semiconductor device has deteriorated.

SUMMARY

[0004] An example embodiment of the present disclosure is to provide a semiconductor device which may increase integration density and may improve performance.

[0005] An example embodiment of the present disclosure is to provide a method of manufacturing the semiconductor device.

[0006] According to an example embodiment, a semiconductor device includes a first structure including a memory region and a first peripheral circuit region; and a second structure overlapping the first structure vertically and including a core circuit region and a second peripheral circuit region, wherein the memory region includes a first cell vertical active pattern; a first word line having a side surface facing a first side surface of the first cell vertical active pattern; and a first cell gate dielectric layer between the first cell vertical active pattern and the first word line, wherein the first peripheral circuit region includes a first peripheral vertical active pattern disposed at substantially the same level as the first cell vertical active pattern; a first peripheral gate electrode having a side surface facing a first side surface of the first peripheral vertical active pattern; and a first peripheral gate dielectric layer between the first peripheral vertical active pattern and the first peripheral gate electrode.

[0007] According to an example embodiment, a semiconductor device includes a first structure including a first memory region and a second memory region spaced apart from each other, and a first peripheral circuit region between the first memory region and the second memory region; and a second structure vertically overlapping and bonded to the first structure, wherein the second structure includes a first core circuit region vertically overlapping the first memory region; a second core circuit region vertically overlapping the second memory region; and a second peripheral circuit region vertically overlapping the first peripheral circuit region, wherein each of the first and second memory regions includes a cell transistor and a data storage structure electrically connected to the cell transistor, wherein the cell transistor includes a cell vertical active pattern; a word line having a side surface facing at least a portion of a side surface of the cell vertical active pattern; and a cell gate dielectric layer between the cell vertical active pattern and the word line.

[0008] According to an example embodiment, a semiconductor device includes a memory region including a cell transistor and data storage structure; a first peripheral circuit region disposed on an outside of the memory region and including the first peripheral circuit transistor; a core circuit region vertically overlapping the memory region and including a core circuit transistor; and a second peripheral circuit region disposed on an outside of the core circuit region, vertically overlapping the first peripheral circuit region, and including a second peripheral circuit transistor.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] Various aspects, features, and advantages of example embodiments of the present disclosure will be more clearly understood from the following detailed description, taken in combination with the accompanying drawings, in which:

[0010] FIGS. **1**A and **1**B are diagrams illustrating a semiconductor device according to an example embodiment of the present disclosure;

[0011] FIG. **2**A is an enlarged diagram illustrating a portion of a semiconductor device according to an example embodiment of the present disclosure;

[0012] FIG. **2**B is an enlarged diagram illustrating a portion of a semiconductor device according to an example embodiment of the present disclosure;

[0013] FIGS. **3**A, **3**B, **3**C, **3**D, **3**E, and **3**F are enlarged cross-sectional diagrams illustrating a portion of a semiconductor device according to an example embodiment of the present disclosure;

[0014] FIGS. **4**A, **4**B, **4**C and **4**D are enlarged cross-sectional diagrams illustrating various examples of transistors in a semiconductor device according to an example embodiment of the present disclosure;

[0015] FIGS. **5**A and **5**B are plan diagrams illustrating various examples of transistors in a semiconductor device according to an example embodiment of the present disclosure;

[0016] FIGS. **6**A and **6**B are diagrams illustrating an inverter in a semiconductor device according to an example embodiment of the present disclosure;

[0017] FIG. **7** is an enlarged cross-sectional diagram illustrating an insulating structure of a semiconductor device according to an example embodiment of the present disclosure;

[0018] FIGS. **8**A, **8**B, **8**C, **8**D, **8**E and **8**F are plan diagrams illustrating various examples of a semiconductor device according to an example embodiment of the present disclosure;

[0019] FIG. **9**A is a cross-sectional diagram illustrating a semiconductor device according to an example embodiment of the present disclosure;

[0020] FIG. **9**B is a cross-sectional diagram illustrating a semiconductor device according to an example embodiment of the present disclosure;

[0021] FIG. **10** is a cross-sectional diagram illustrating a semiconductor device according to an example embodiment of the present disclosure;

[0022] FIG. **11** is a cross-sectional diagram illustrating a semiconductor device according to an example embodiment of the present disclosure;

[0023] FIG. **12** is a cross-sectional diagram illustrating a semiconductor device according to an example embodiment of the present disclosure;

[0024] FIG. **13** is a cross-sectional diagram illustrating a semiconductor device according to an example embodiment of the present disclosure;

[0025] FIG. **14** is a perspective diagram illustrating a semiconductor device according to an example embodiment of the present disclosure;

[0026] FIG. **15** is a flowchart illustrating a method of manufacturing a semiconductor device according to an example embodiment of the present disclosure; and

[0027] FIGS. **16**A to **16**H and **17** are cross-sectional diagrams illustrating a method of manufacturing a semiconductor device according to an example embodiment of the present disclosure.

DETAILED DESCRIPTION

[0028] Hereinafter, example embodiments of the present disclosure will be described as follows with reference to the accompanying drawings. Hereinafter, terms such as “upper,” “intermediate,” “middle,” “lower” and the like may be replaced with other terms, such as “first,” “second,” “third” and the like to describe the elements of the specification. Terms such as “first,” “second” and “third” may be used to describe various elements, but the elements are not limited in order or relative position by the terms, and an element referred to as a “first element” may be equivalently referred to as a “second element.” In the specification, terms such as ‘lower’, ‘upper’, ‘upper end’, and ‘lower end’ may be terms that are described with reference to the drawings. A vertical length of the first element may refer to the distance from the lower surface to the upper surface of the first element.

[0029] An example of a semiconductor device according to an example embodiment will be described with reference to FIGS. **1**A and **1**B. FIG. **1**A is a cross-sectional diagram illustrating a semiconductor device according to an example embodiment, and FIG. **1**B is an enlarged diagram illustrating regions “A” and “B” in FIG. **1**A.

[0030] Referring to FIGS. **1**A and **1**B, a semiconductor device **1** according to an example embodiment may include a first structure CH**1** and a second structure CH**2** overlapping the first structure vertically CH**1**. The first structure CH**1** may be disposed on the second structure CH**2**. The first structure CH**1** may be bonded to the second structure CH**2**.

[0031] The first structure CH**1** may include a first memory region MR**1** and a first peripheral circuit region PR_U disposed on an outside of the first memory region MR**1**.

[0032] The first structure CH**1** may further include a second memory region MR**2** disposed at the same level as a level of the first memory region MR**1**.

[0033] The first peripheral circuit region PR_U may be disposed between the first memory region MR**1** and the second memory region MR**2**. The first memory region MR**1** and the second memory region MR**2** may have substantially the same structure.

[0034] Each of the first and second memory regions MR**1** and MR**2** may include cell transistors c_TRa and c_TRb. For example, the first memory region MR**1** may include first cell transistors c_TRa, and the second memory region MR**2** may include second cell transistors c_TRb. The cell transistors c_TRa and c_TRb may also be referred to as cell vertical channel transistors.

[0035] The first memory region MR**1** may include cell vertical active patterns **109***m***1**, word lines **139***m***1**, and cell gate dielectric layers FOc between the cell vertical active patterns **109***m***1** and the word lines **139***m***1** included in the first cell transistors c_TRa.

[0036] The cell vertical active patterns **109***m***1** may include a first cell vertical active pattern **109***m***1***a* and a second cell vertical active pattern **109***m***1***b* adjacent to and spaced apart from each other.

[0037] The word lines **139***m***1** may include a first word line **139***m***1***a* having a side surface facing a first side surface of the first cell vertical active pattern **109***m***1***a*, and a second word line **139***m***1***b* having a side surface facing a second side surface of the second cell vertical active pattern **109***m***1***b*.

[0038] The cell gate dielectric layers FOc may be disposed between the first cell vertical active pattern **109***m***1***a* and the first word line **139***m***1***a*, and between the second cell vertical active pattern **109***m***1***b* and the second word line **139***m***1***b*.

[0039] The first memory region MR**1** may further include cell back gate electrodes **121***m***1**, and cell back gate dielectric layers BOc between the cell back gate electrodes **121***m***1** and the cell vertical active patterns **109***m***1**. For example, the cell back gate electrode **121***m***1** may be disposed between the first cell vertical active pattern **109***m***1***a* and the second cell vertical active pattern **109***m***1***b* adjacent to each other, and the cell back gate dielectric layers BOc may be disposed between the first cell vertical active pattern **109***m***1***a* and the cell back gate electrode **121***m***1**, and between the second cell vertical active pattern **109***m***1***b* and the cell back gate electrode **121***m***1**.

[0040] The first word line **139***m***1***a* may have a side surface facing the first side surface of the first cell vertical active pattern **109***m***1***a*, the first word line **139***m***1***b* may have a side surface facing the second side surface of the second cell vertical active pattern **109***m***1***b*, and the cell back gate electrode **121***m***1** may be disposed between a third side surface of the first cell vertical active pattern **109***m***1***a* and a fourth side surface of the second cell vertical active pattern **109***m***1***b*, with the third side surface and the fourth side surface facing each other.
[0041] The first cell vertical active pattern **109***m***1***a*, the cell back gate electrode **121***m***1**, and the second cell vertical active pattern **109***m***1***b* may be disposed between the first word line **139***m***1***a* and the second word line **139***m***1***b*.
[0042] Each of the cell vertical active patterns **109***m***1** may include a lower cell source/drain region **148***m*, an upper cell source/drain region **187***m* on the lower cell source/drain region **148***m*, and a cell channel region CHm between the lower cell source/drain region **148***m* and the upper cell source/drain region **187***m*.
[0043] The lower cell source/drain region **148***m* and the upper cell source/drain region **187***m* may have N-type conductivity.
[0044] The first word line **139***m***1***a* may have a side surface facing the cell channel region CHm of the first cell vertical active pattern **109***m***1***a*. The first word line **139***m***1***b* may have a side surface facing the cell channel region CHm of the second cell vertical active pattern **109***m***1***b*.
[0045] The second memory region MR**2** and the first memory region MR**1** may have substantially the same structure, such that the second memory region MR**2** may include cell vertical active patterns **109***m***2**, word lines **139***m***2** and cell back gate electrodes **121***m***2**, corresponding to the first memory region MR**1**custom-character the cell vertical active patterns **109***m***1**, the word lines **139***m***1** and the cell back gate electrodes **121***m***1**, respectively.
[0046] The cell back gate electrodes **121***m***1** and **121***m***2** may control charges accumulated in the cell channel regions CHm of the cell vertical active patterns **109***m***1** and **109***m***2**. Accordingly, the cell channel regions CHm may be configured as a floating body disposed between the lower and upper cell source/drain regions **148***m* and **187***m*, and the cell back gate electrodes **121***m***1** and **121***m***2** may prevent performance of the cell transistors c_TRa and c_TRb from being deteriorated due to the floating body effect and may improve performance of the cell transistors c_TRa and c_TRb. Accordingly, the cell back gate electrodes **121***m***1** and **121***m***2** may prevent charges, for example, holes from being accumulated in the floating body of the cell channel regions CHm and changing a threshold voltage of the cell transistors c_TRa and c_TRb during operation of the cell transistors c_TRa and c_TRb. Accordingly, the cell back gate electrodes **121***m***1** and **121***m***2** may allow the cell transistors c_TRa and c_TRb to operate stably, and performance of the semiconductor device **1** may be improved.
[0047] The first peripheral circuit region PR_U may include the first peripheral circuit transistors N_TRa and P_TRa disposed at substantially the same level as the cell transistors c_TRa and c_TRb. For example, the first peripheral circuit region PR_U may include first peripheral NMOS transistors N_TRa and first peripheral PMOS transistors P_PRa.
[0048] The first peripheral circuit region PR_U may include peripheral gate electrodes **139***p***1** and **139***p***2**, peripheral vertical active patterns **109***p***1** and **109***p***2**, and peripheral gate dielectric layers FOn and FOp included in the first peripheral circuit transistors N_TRa and P_TRa. The peripheral vertical active patterns **109***p***1** and **109***p***2** may include NMOS vertical active patterns **109***p***1** and PMOS vertical active patterns **109***p***2**.
[0049] The cell vertical active patterns **109***m***1** and **109***m***2** and the peripheral vertical active patterns **109***p***1** and **109***p***2** may include a material used as a channel of a transistor, for example, a semiconductor material. For example, each of the cell vertical active patterns **109***m***1** and **109***m***2** and the peripheral vertical active patterns **109***p***1** and **109***p***2** may include at least one of a silicon layer, oxide semiconductor layer, and two-dimensional material layer having semiconductor properties, which may be used as a channel region. For example, each of the cell vertical active

patterns **109**_m_**1** and **109**_m_**2** and the peripheral vertical active patterns **109**_p_**1** and **109**_p_**2** may include single crystal silicon or polysilicon.

[0050] The peripheral gate electrodes **139**_p_**1** and **139**_p_**2** may include NMOS gate electrodes **139**_p_**1** and PMOS gate electrodes **139**_p_**2**. The peripheral gate dielectric layers FOn and FOp may include NMOS gate dielectric layers FOn and PMOS gate dielectric layers FOp.

[0051] The word lines **139**_m_**1** and **139**_m_**2**, the NMOS gate electrodes **139**_p_**1**, and the PMOS gate electrodes **139**_p_**2** may be formed of doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, or a combination thereof, but an example embodiment thereof is not limited thereto. Each of the word lines **139**_m_**1** and **139**_m_**2**, the NMOS gate electrodes **139**_p_**1**, and the PMOS gate electrodes **139**_p_**2** may include a single layer or multiple layers formed of the aforementioned conductive materials. In an example embodiment, the NMOS gate electrodes **139**_p_**1** and the PMOS gate electrodes **139**_p_**2** may include conductive materials having different work functions.

[0052] The NMOS vertical active patterns **109**_p_**1** may include a first NMOS vertical active pattern **109**_p_**1**_a_ and a second NMOS vertical active pattern **109**_p_**1**_b_ adjacent to and spaced apart from each other.

[0053] In an example, a spacing distance between the cell vertical active patterns **109**_m_**1** adjacent to each other with the cell back gate electrode **121**_m_**1** therebetween may be substantially the same as a spacing distance between the first NMOS vertical active pattern **109**_p_**1**_a_ and the second NMOS vertical active pattern **109**_p_**1**_b_ adjacent to each other.

[0054] The NMOS gate electrodes **139**_p_**1** may have a first NMOS gate electrode **139**_p_**1**_a_ having a side surface facing the first side surface of the first NMOS vertical active pattern **109**_p_**1**_a_, and a second NMOS gate electrode **139**_p_**1**_b_ having a side surface facing the second side surface of the second NMOS vertical active pattern **109**_p_**1**_b_.

[0055] The NMOS gate dielectric layers FOn may be disposed between the first NMOS vertical active pattern **109**_p_**1**_a_ and the first NMOS gate electrode **139**_p_**1**_a_, and between the second NMOS vertical active pattern **109**_p_**1**_b_ and the second NMOS gate electrode **139**_p_**1**_b_.

[0056] The PMOS vertical active patterns **109**_p_**2** may include a first PMOS vertical active pattern **109**_p_**2**_a_ and a second PMOS vertical active pattern **109**_p_**2**_b_ adjacent to and spaced apart from each other.

[0057] In an example, a spacing distance between the cell vertical active patterns **109**_m_**1** adjacent to each other with the cell back gate electrode **121**_m_**1** therebetween may be substantially the same as a spacing distance between the first PMOS vertical active pattern **109**_p_**2**_a_ and the second PMOS vertical active pattern **109**_p_**2**_b_ adjacent to each other.

[0058] The PMOS gate electrodes **139**_p_**2** may have a first PMOS gate electrode **139**_p_**2**_a_ having a side surface facing the first side surface of the first PMOS vertical active pattern **109**_p_**2**_a_, and a second PMOS gate electrode **139**_p_**2**_b_ having a side surface facing the second side surface of the second PMOS vertical active pattern **109**_p_**2**_b_.

[0059] The PMOS gate dielectric layers FOp may be disposed between the first PMOS vertical active pattern **109**_p_**2**_a_ and the first PMOS gate electrode **139**_p_**2**_a_, and between the second PMOS vertical active pattern **109**_p_**2**_b_ and the second PMOS gate electrode **139**_p_**2**_b_.

[0060] The NMOS gate electrodes **139**_p_**1** may have a vertical length H1 substantially the same as that of the word lines **139**_m_**1** and **139**_m_**2**.

[0061] The PMOS gate electrodes **139**_p_**2** may have a vertical length H1 substantially the same as that of the word lines **139**_m_**1** and **139**_m_**2**.

[0062] The first peripheral circuit region PR_U may further include third peripheral gate electrodes **121**_p_**1** and **121**_p_**2** and third peripheral gate dielectric layers BOn and BOp.

[0063] The third peripheral gate electrodes **121**_p_**1** and **121**_p_**2** may be configured as peripheral back gate electrodes. Similarly to the cell back gate electrodes **121**_m_**1** and **121**_m_**2**, the third peripheral gate electrodes **121**_p_**1** and **121**_p_**2**, which may be peripheral back gate electrodes, may enable the

peripheral circuit transistors N_TRa and P_TR to operate stably, such that performance of the semiconductor device **1** may be improved.
[0064] The cell back gate electrodes **121***m***1** and **121***m***2** and the third peripheral gate electrodes **121***p***1** and **121***p***2** may include at least one conductive material. For example, each of the cell back gate electrodes **121***m***1** and **121***m***2** and the third peripheral gate electrodes **121***p***1** and **121***p***2** may be formed of doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, or a combination thereof, but an example embodiment thereof is not limited thereto. Each of the cell back gate electrodes **121***m***1** and **121***m***2** and the third peripheral gate electrodes **121***p***1** and **121***p***2** may include a single layer or multiple layers formed of the aforementioned materials.
[0065] In an example, the cell back gate electrodes **121***m***1** and **121***m***2** and the third peripheral gate electrodes **121***p***1** and **121***p***2** may include a material different from that of the word lines **139***m***1** and **139***m***2**.
[0066] In another example, the cell back gate electrodes **121***m***1** and **121***m***2** and the third peripheral gate electrodes **121***p***1** and **121***p***2** may include the same material as that of the word lines **139***m***1** and **139***m***2**.
[0067] In an example, the cell back gate electrodes **121***m***1** and **121***m***2** and the third peripheral gate electrodes **121***p***1** and **121***p***2** may have the same vertical length.
[0068] In an example, the vertical length of the cell back gate electrodes **121***m***1** and **121***m***2** and the third peripheral gate electrodes **121***p***1** and **121***p***2** may be different from the vertical length of the word lines **139***m***1** and **139***m***2** and the peripheral gate electrodes **139***p***1** and **139***p***2**, respectively.
[0069] In another example, the vertical length of the cell back gate electrodes **121***m***1** and **121***m***2** and the third peripheral gate electrodes **121***p***1** and **121***p***2** may be the same as the vertical length of the word lines **139***m***1** and **139***m***2** and the peripheral gate electrodes **139***p***1** and **139***p***2**, respectively.
[0070] The third peripheral gate electrodes **121***p***1** and **121***p***2** may include a third NMOS gate electrode **121***p***1** disposed between the first NMOS vertical active pattern **109***p***1***a* and the second NMOS vertical active pattern **109***p***1***b*, adjacent to each other, and a third PMOS gate electrode **121***p***2** disposed between the first PMOS vertical active pattern **109***p***2***a* and the second PMOS vertical active pattern **109***p***2***b*, adjacent to each other.
[0071] The third peripheral gate dielectric layers BOn and BOp may include third NMOS gate dielectric layers BOn and third PMOS gate dielectric layers BOp.
[0072] The third NMOS gate dielectric layers BOn may be disposed between the first NMOS vertical active pattern **109***p***1***a* and the third NMOS gate electrode **121***p***1**, and between the second NMOS vertical active pattern **109***p***1***b* and the third NMOS gate electrode **121***p***1**.
[0073] The third PMOS gate dielectric layers BOp may be disposed between the first PMOS vertical active pattern **109***p***2***a* and the third PMOS gate electrode **121***p***2**, and between the second PMOS vertical active pattern **109***p***2***b* and the third PMOS gate electrode **121***p***2**.
[0074] The first NMOS gate electrode **139***p***1***a* may have a side surface facing the first side surface of the first NMOS vertical active pattern **109***p***1***a*, the second NMOS gate electrode **139***p***1***b* may have a side surface facing the second side surface of the second NMOS vertical active pattern **109***p***1***b*, and the third NMOS gate electrode **121***p***1** may be disposed between the third side surface of the first NMOS vertical active pattern **109***p***1***a* and the fourth side surface of the second NMOS vertical active pattern **109***p***1***b*, with the third side surface and the fourth side surface facing each other.
[0075] The first NMOS vertical active pattern **109***p***1***a*, the third NMOS gate electrode **121***p***1**, and the second NMOS vertical active pattern **109***p***1***b* may be disposed between the first NMOS gate electrode **139***p***1***a* and the second NMOS gate electrode **139***p***1***b*.
[0076] In an example embodiment, the third NMOS gate electrode **121***p***1** may be configured as an NMOS back gate electrode.
[0077] The first PMOS gate electrode **139***p***2***a* may have a side surface facing the first side surface

of the first PMOS vertical active pattern **109***p***2***a*, the second PMOS gate electrode **139***p***2***b* may have a side surface facing the second side surface of the second PMOS vertical active pattern **109***p***2***b*, and the third PMOS gate electrode **121***p***2** may be disposed between the third side surface of the first PMOS vertical active pattern **109***p***2***a* and the fourth side surface of the second PMOS vertical active pattern **109***p***2***b*, with the third side surface and the fourth side surface facing each other.

[0078] The first PMOS vertical active pattern **109***p***2***a*, the third PMOS gate electrode **121***p***2**, and the second PMOS vertical active pattern **109***p***2***b* may be disposed between the first PMOS gate electrode **139***p***2***a* and the second PMOS gate electrode **139***p***2***b*.

[0079] In an example embodiment, the third PMOS gate electrode **121***p***2** may be configured as a PMOS back gate electrode.

[0080] Each of the NMOS vertical active patterns **109***p***1** may include a lower NMOS source/drain region **148***p***1**, an upper NMOS source/drain region **187***p***1** on the lower NMOS source/drain region **148***p***1**, and an NMOS channel region CHp***1*** between the lower NMOS source/drain region **148***p***1** and the upper NMOS source/drain region **187***p***1**.

[0081] The lower NMOS source/drain region **148***p***1** and the upper NMOS source/drain region **187***p***1** may have N-type conductivity.

[0082] In an example, the lower NMOS source/drain region **148***p***1** and the upper NMOS source/drain region **187***p***1** may have an impurity concentration substantially the same as those of the lower cell source/drain region **148***m* and the upper cell source/drain region **187***m*.

[0083] In another example, the lower NMOS source/drain region **148***p***1** and the upper NMOS source/drain region **187***p***1** may have an impurity concentration higher than impurity concentrations of the lower cell source/drain region **148***m* and the upper cell source/drain region **187***m*.

[0084] Each of the PMOS vertical active patterns **109***p***2** may include a lower PMOS source/drain region **148***p***2**, an upper PMOS source/drain region **187***p***2** on the lower PMOS source/drain region **148***p***2**, and a PMOS channel region CHp**2** between the lower PMOS source/drain region **148***p***2** and the upper PMOS source/drain region **187***p***2**.

[0085] The lower PMOS source/drain region **148***p***2** and the upper PMOS source/drain region **187***p***2** may have P-type conductivity.

[0086] The first NMOS gate electrode **139***p***1***a* may have a side surface facing the NMOS channel region CHp**1** of the first NMOS vertical active pattern **109***p***1***a*. The second NMOS gate electrode **109***p***1***b* may have a side surface facing the NMOS channel region CHp**1** of the second NMOS vertical active pattern **109***p***1***b*.

[0087] The first PMOS gate electrode **139***p***2***a* may have a side surface facing the PMOS channel region CHp**2** of the first PMOS vertical active pattern **109***p***2***a*. The second PMOS gate electrode **139***p***2***b* may have a side surface facing the PMOS channel region CHp**2** of the second PMOS vertical active pattern **109***p***2***b*.

[0088] Each of the first and second memory regions MR**1** and MR**2** may further include data storage structures DS**1** and DS**2**.

[0089] The data storage structures DS**1** and DS**2** may be disposed on the cell transistors c_TRa and c_TRb. The data storage structures DS**1** and DS**2** may be electrically connected to the cell transistors c_TRa and c_TRb. The data storage structures DS**1** and DS**2** may include the first data storage structure DS**1** in the first memory region MR**1** and the second data storage structure DS**2** in the second memory region MR**2**.

[0090] The data storage structures DS**1** and DS**2** may be configured as memory cell capacitors which may store data in a memory such as a DRAM, but an example embodiment thereof is not limited thereto. For example, the data storage structures DS**1** and DS**2** may be configured as a data storage structure of MRAM or a data storage structure of FeRAM.

[0091] Each of the data storage structures DS**1** and DS**2** may include first electrodes **163** extending in the vertical direction Z, a second electrode **169** on a side surface and an upper surface of each of the first electrodes **163**, and a dielectric layer **166** between the first electrodes **163** and the second electrode **169**.

[0092] Each of the first and second memory regions MR**1** and MR**2** may further include cell contact structures **161***a* and **161***b* between the cell transistors c_TRa and c_TRb and the data storage structures DS**1** and DS**2**. The cell contact structures **16***a* and **161***b* may be disposed on the cell transistors c_TRa and c_TRb. The cell contact structures **161***a* and **161***b* may be electrically connected to the upper cell source/drain regions **187***m* of the cell transistors c_TRa and c_TRb.

[0093] The cell contact structures **161***a* and **161***b* may include a first cell contact structures **161***a* which may electrically connect the first cell transistors c_TRa to the first data storage structure DS**1** in the first memory region MR**1**, and a second cell contact structures **161***b* which may electrically connect the second cell transistors c_TRb to the second data storage structure DS**2** in the second memory region MR**2**.

[0094] Each of the first and second cell contact structures **161***a* and **161***b* may include a cell plug pattern **154***m***1**, **154***m***2**, and a cell pad pattern **160***m***1**, **160***m***2** on the cell plug pattern **154***m***1**, **154***m***2**. A vertical central axis of the cell pad pattern **160***m***1**, **160***m***2** may not be aligned with a vertical central axis of the cell plug pattern **154***m***1**, **154***m***2**.

[0095] The cell plug patterns **154***m***1**, **154***m***2** may be connected to the upper cell source/drain regions **187***m*. The cell pad patterns **160***m***1**, **160***m***2** may be connected to the first electrodes **163**.

[0096] The first peripheral circuit region PR_U may further include a peripheral vertical active patterns **109***p***1** and **109***p***2** on the peripheral contact structures **154***p***1** and **154***p***2** and a peripheral contact structures **154***p***1** and **154***p***2** on the upper peripheral wirings **160***pa* and **160***pb*.

[0097] The peripheral contact structures **154***p***1** and **154***p***2** may be disposed at substantially the same level as the cell plug patterns **154***m***1**. The peripheral contact structures **154***p***1** and **154***p***2** and the cell plug patterns **154***m***1** may include the same conductive material.

[0098] The peripheral contact structures **154***p***1** and **154***p***2** may include NMOS contact structures **154***p***1** electrically connected to the upper NMOS source/drain regions **187***p***1** and PMOS contact structures **154***p***2** electrically connected to the upper PMOS source/drain regions **187***p***2**.

[0099] The upper peripheral wirings **160***pa* and **160***pb* may be disposed at substantially the same level as the cell pad patterns **160***m***1**. The upper peripheral wirings **160***pa* and **160***pb* and the cell pad patterns **160***m***1** may include the same conductive material.

[0100] The upper peripheral wirings **160***pa* and **160***pb* may include first upper peripheral wirings **160***pa* and second upper peripheral wirings **160***pb*.

[0101] The first memory region MR**1** may further include a first bit line **190***m***1** below the cell transistors c_TRa, and the second memory region MR**2** may further include a second bit line **190***m***2** below the cell transistors c_TRb.

[0102] The first bit line **190***m***1** may be connected to the lower cell source/drain regions **148***m* of the cell vertical active patterns **109***m***1** in the first memory region MR**1**, and the second bit line **190***m***2** may be connected to the lower cell source/drain regions **148***m* of the cell vertical active patterns **109***m***2** in the second memory region MR**2**.

[0103] The first peripheral circuit region PR_U may further include lower peripheral wirings **190***p***1** and **190***p***2** disposed at substantially the same level as the first and second bit lines **190***m***1** and **190***m***2** and disposed below the peripheral circuit transistors N_TRa and P_TRa.

[0104] The lower peripheral wirings **190***p***1** and **190***p***2** may include a first lower peripheral wiring **190***p***1** connected to the NMOS vertical active patterns **109***p***1** and a second lower peripheral wiring **190***p***2** connected to the PMOS vertical active patterns **109***p***2**.

[0105] Each of the bit lines **190***m*and **190***m***2** and the lower peripheral wirings **190***p***1** and **190***p***2** may include at least one conductive material layer. For example, each of the bit lines **190***m***1** and **190***m***2** and the lower peripheral wirings **190***p***1** and **190***p***2** may include doped polysilicon, metal, conductive metal nitride, metal-semiconductor compound, conductive metal oxide, conductive graphene, and carbon nanotube or a combination thereof. For example, each of the bit lines **190***m***1**

and **190***m***2** and the lower peripheral wirings **190***p***1** and **190***p***2** may be formed of doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrOx, RuOx, graphene, carbon nanotube, or a combination thereof, but an example embodiment thereof is not limited thereto. Each of the bit lines **190***m***1** and **190***m***2** and the lower peripheral wirings **190***p***1** and **190***p***2** may include a single layer or multiple layers formed of the aforementioned conductive materials. For example, each of the bit lines **190***m***1** and **190***m***2** and the lower peripheral wirings **190***p***1** and **190***p***2** may include a first conductive layer **190***a* and a second conductive layer **190***b* below the first conductive layer **190***a*.
[0106] The first conductive layer **190***a* of the first lower peripheral wiring **190***p***1** may include polysilicon having N-type conductivity, and the first conductive layer **190***a* of the second lower peripheral wiring **190***p***2** may include polysilicon having P-type conductivity.
[0107] The first structure CH**1** may further include a first routing wiring structure **192** disposed on a level lower than levels of the bit lines **190***m***1** and **190***m***2** and the lower peripheral wirings **190***p***1** and **190***p***2**, and first bonding pads **196** disposed on a level lower than a level of the first routing wiring structure **192**.
[0108] The first routing wiring structure **192** may include a plurality of wirings disposed on different levels and a plurality of vias connected to an upper surface and a lower surface of the plurality of wirings, respectively.
[0109] The first routing wiring structure **192** may be electrically connected to the bit lines **190***m***1** and **190***m***2**. In example embodiments, the first routing wiring structure **192** may be electrically connected to the lower peripheral wirings **190***p***1** and **190***p***2**.
[0110] The first structure CH**1** may further include connection contact plugs **194** disposed between the first routing wiring structure **192** and the first upper peripheral wirings **160***pa* and electrically connecting the first routing wiring structure **192** to the first upper peripheral wirings **160***pa*.
[0111] The connection contact plugs **194** may be in contact with lower surfaces of the first upper peripheral wirings **160***pa*, may extend downwardly and may be connected to one of the plurality of wirings of the first routing wiring structure **192**.
[0112] The first structure CH**1** may include first, second, third, fourth, and fifth insulating structures **198**, INS, **151**, **157**, and **175**.
[0113] The first insulating structure **198** may have a lower surface coplanar with the lower surfaces of the first bonding pads **196**, and may have an upper surface coplanar with upper surfaces of the bit lines **190***m***1** and **190***m***2** and the lower peripheral wirings **190***p***1** and **190***p***2**. The first insulating structure **198** may include a plurality of insulating layers.
[0114] The second insulating structure INS may have a lower surface coplanar with lower surfaces of the cell vertical active patterns **109***m***1** and **109***m***2**, the NMOS vertical active patterns **109***p***1**, and the PMOS vertical active patterns **109***p***2**, and an upper surface coplanar with upper surfaces of the cell vertical active patterns **109***m***1** and **109***m***2**, the NMOS vertical active patterns **109***p***1**, and the PMOS vertical active patterns **109***p***2**. The second insulating structure INS may include a plurality of insulating layers.
[0115] The third insulating structure **151** may be disposed on side surfaces of the cell plug patterns **154***m***1** and the peripheral contact structures **154***p***1** and **154***p***2**.
[0116] The fourth insulating structure **157** may be disposed on side surfaces of the cell pad patterns of **160***m***1** and the upper peripheral wirings of **160***pa* and **160***pb*.
[0117] The fifth insulating structure **175** may be disposed on the data storage structures DS**1** and DS**2** and the upper peripheral wirings **160***pa* and **160***pb*.
[0118] The first structure CH**1** may further include cell contact plugs **178***m* on the second electrodes **169** of the data storage structures DS**1** and DS**2** and peripheral contact plug **178***p* on the second upper peripheral wiring **160***pb*.
[0119] The cell contact plugs **178***m* may penetrate the fifth insulating structure **175** and may be electrically connected to the second electrodes **169** of the data storage structures DS**1** and DS**2**.

[0120] The peripheral contact plug **178***p* may penetrate the fifth insulating structure **175** and may be electrically connected to the second upper peripheral wiring **160***pb*.
[0121] The first structure CH**1** may further include the cell upper wirings **184***m* on the cell contact plugs **178***m* and the third peripheral upper wiring **184***p* on the peripheral contact plug **178***p*. The cell upper wirings **184***m* and the third peripheral upper wiring **184***p* may be disposed on substantially the same level.
[0122] The first structure CH**1** may further include an upper insulating structure **181** on the fifth insulating structure **175**, the cell upper wirings **184***m*, and the third peripheral upper wiring **184***p*.
[0123] The second structure CH**2** may include a first core circuit region CR**1** vertically overlapping the first memory region MR**1** and a second peripheral circuit region PR_L vertically overlapping the first peripheral circuit region PR_U.
[0124] The second structure CH**2** may further include a second core circuit region CR**2** vertically overlapping the second memory region MR**2**. The second peripheral circuit region PR_L may be disposed between the first core circuit region CR**1** and the second core circuit region CR**2**.
[0125] The second structure CH**2** may include a substrate **5** and a device isolation region **10***s* limiting active regions **10***a* on the substrate **5**. The substrate **5** may be configured as a semiconductor substrate.
[0126] The second structure CH**2** may further include lower peripheral circuit transistors **25***a*, **25***b*, and **25***c* disposed on the substrate **5**, a second routing wiring structure **30** electrically connected to the lower peripheral circuit transistors **25***a*, **25***b*, and **25***c* on the lower peripheral circuit transistors **25***a*, **25***b*, and **25***c*, second bonding pads **35** electrically connected to the second routing wiring structure **30** on the second routing wiring structure **30**, and a lower insulating structure **40**.
[0127] Each of the lower peripheral circuit transistors **25***a*, **25***b*, and **25***c* may include peripheral gate structures **15***a* and **15***b* disposed on the active region **10***a*, and peripheral source/drain regions **20***sd* disposed in the active region **10***a* disposed on both sides of the peripheral gate structure **15***a* and **15***b*. The peripheral gate structures **15***a* and **15***b* may include peripheral gate dielectric layer **15***a* and peripheral gate electrode **15***b* stacked in order.
[0128] The lower peripheral circuit transistors **25***a*, **25***b*, and **25***c* may include a first core circuit transistor **25***a* in the first core circuit region CR**1**, a second core circuit transistor **25***b* in the second core circuit region CR**2**, and a second peripheral circuit transistor **25***c* in the second peripheral circuit region PR_L.
[0129] The first core circuit region CR**1** may include the first core circuit transistor **25***a*. The first core circuit transistor **25***a* may be electrically connected to the first bit line **190***m***1** through the second routing wiring structure **30**, the second bonding pad **35**, the first bonding pad **196** and the first routing wiring structure **192**.
[0130] The second core circuit region CR**2** may include the second core circuit transistor **25***b*. The second core circuit transistor **25***b* may be electrically connected to the second bit line **190***m***2** through the second routing wiring structure **30**, the second bonding pad **35**, the first bonding pad **196** and the first routing wiring structure **192**.
[0131] The first core circuit transistor **25***a* may vertically overlap the first data storage structure DS**1**, and the second core circuit transistor **25***b* may vertically overlap the second data storage structure DS**2**.
[0132] The first peripheral circuit region PR_U and the second peripheral circuit region PR_L may not vertically overlap the first and second data storage structures DS**1** and DS**2**.
[0133] The lower insulating structure **40** may be disposed on the substrate **5** and may have an upper surface coplanar with upper surfaces of the second bonding pads **35**.
[0134] Upper surfaces of the second bonding pads **35** may be bonded to lower surfaces of the first bonding pads **196**, and upper surfaces of the lower insulating structure **40** may be bonded to lower surfaces of the first insulating structure **198**. Accordingly, a bonding surface JSa may be formed between the first structure CH**1** and the second structure CH**2**.

[0135] In the description below, an example of a semiconductor device according to an example embodiment will be described with reference to FIG. **2**A. FIG. **2**A is an enlarged diagram illustrating region A of FIG. **1**B and further illustrating region B′ that may replace the region B in FIG. **1**B.

[0136] Referring to the region A, the cell vertical active patterns **109***m***1** described above, adjacent to each other with the cell back gate electrode **121***m***1** therebetween, may be spaced apart each other from with a first distance D**1**.

[0137] In an example, the first NMOS vertical active pattern (**109***p***1***a* in FIG. **1**B) and the second NMOS vertical active pattern (**109***p***1***b* in FIG. **1**B) described above, adjacent to each other, may be replaced with a first NMOS vertical active pattern **109***p***1***a′* and a second NMOS vertical active pattern **109***p***1***b′* spaced apart from each other with a second distance D**2**. The second distance D**2** may be different from the first distance D**1**. The second distance D**2** may be greater than the first distance D**1**.

[0138] In an example, the first PMOS vertical active pattern (**109***p***2***a* in FIG. **1**B) and the second PMOS vertical active pattern (**109***p***2***b* in FIG. **1**B) described above, adjacent to each other, may be replaced with a first PMOS vertical active pattern **109***p***2***a′* and a second PMOS vertical active pattern **109***p***2***b′* spaced apart from each other with the second distance D**2**.

[0139] In an example, the third NMOS gate electrode (**121***p***1** in FIG. **1**B) described above, disposed between the first NMOS vertical active pattern (**109***p***1***a* in FIG. **1**B) and the second NMOS vertical active pattern (**109***p***1***b* in FIG. **1**B) adjacent to each other may be replaced with a third NMOS gate electrode **121***p***1***a* and a fourth NMOS gate electrode **121***p***1***b* spaced apart from each other in the first NMOS vertical active pattern **109***p***1***a′* and the second NMOS vertical active pattern **109***p***1***b′*.

[0140] In an example, the third and fourth NMOS gate electrodes **121***p***1***a* and **121***p***1***b* may be the peripheral back gate electrode described above.

[0141] In an example, the same bias as that of the first and second NMOS gate electrodes **139***p***1***a* and **139***p***1***b* may be applied to the third and fourth NMOS gate electrodes **121***p***1***a* and **121***p***1***b*. Accordingly, first NMOS transistors N_TRa′ using the third and fourth NMOS gate electrodes **121***p***1***a* and **121***p***1***b* and the first and second NMOS gate electrodes **139***p***1***a* and **139***p***1***b* as NMOS gate electrodes may be provide. For example, the first NMOS transistor N_TRa′ may include the first NMOS gate electrode **139***l***1***a* and the third NMOS gate electrode **121***p***1***a* applied with the same bias to work as an NMOS gate electrode on both sides of the first NMOS vertical active pattern **109***p***1***a*.

[0142] In an example, the third PMOS gate electrode (**121***p***2** in FIG. **1**B) described above, disposed between the first PMOS vertical active pattern (**109***p***2***a* in FIG. **1**B) and the second PMOS vertical active pattern (**109***p***2***b* in FIG. **1**B) adjacent to each other may be replaced with a third PMOS gate electrode **121***p***2***a* and a fourth PMOS gate electrode **121***p***2***b* spaced apart from each other in the first PMOS vertical active pattern **109***p***2***a′* and the second PMOS vertical active pattern **109***p***2***b′*.

[0143] In an example, the third and fourth PMOS gate electrodes **121***p***2***a* and **121***p***2***b* may be the peripheral back gate electrodes described above.

[0144] In an example, the same bias as the first and second PMOS gate electrodes **139***p***2***a* and **139***p***2***b* may be applied to the third and fourth PMOS gate electrodes **121***p***2***a* and **121***p***2***b*. Accordingly, the first PMOS transistors P_TRa′ using the third and fourth PMOS gate electrodes **121***p***2***a* and **121***p***2***b* and the first and second PMOS gate electrodes **139***p***2***a* and **139***p***2***b* as PMOS gate electrodes may be provided. For example, the first PMOS transistor P_TRa′ may include the first PMOS gate electrode **139***p***2***a* and the third PMOS gate electrode **121***p***2***a* applied with the same bias to work as a PMOS gate electrode on both sides of the first PMOS vertical active pattern **109***p***2***a*.

[0145] In the description below, an example of a semiconductor device according to an example embodiment will be described with reference to regions A′ and B″ in FIG. **2**B. The regions A′ and B″ are enlarged diagrams illustrating regions that may replace regions “A” and “B” in FIGS. **1**A and **1**B.

[0146] Referring to A′ and B″, the cell back gate electrode **121***m***1** and the cell back gate dielectric layer BOc described with reference to FIGS. **1**A and **1**B may not be provided. The third NMOS gate electrode **121***p***1**, the third NMOS gate dielectric layer BOn, the third PMOS gate electrode **121***p***2**, and the third PMOS gate dielectric layer BOp described with reference to FIGS. **1**A and **1**B may not be provided.

[0147] In the description below, example embodiments of various types of first peripheral circuit transistors or peripheral contact plugs included in the first peripheral circuit region PR_U described above will be described with reference to FIGS. **3**A to **3**F. In FIGS. **3**A to **3**F, region B**1** may correspond to region B**1** in FIG. **1**B, and region B**2** may correspond to region B**2** in FIG. **1**B. For example, in FIGS. **3**A to **3**F, region B**1** may be a region in which a peripheral NMOS transistor is disposed, and region B**2** may be a region in which a peripheral PMOS transistor is disposed. In FIGS. **3**A to **3**F, the same reference numeral as in FIG. **1**B may indicate an element having the same form as described in FIG. **1**B. Accordingly, in FIGS. **3**A to **3**F, the elements modified or replaced with different shapes from the elements in FIG. **1**B will be described.

[0148] In an example, referring to FIG. **3**A, a peripheral NMOS gate electrode **139***p***11** having a vertical length H**2** smaller than the vertical length H**1** of the word lines (**139***m***1** in FIG. **1**B) may be provided. A peripheral PMOS gate electrode **139***p***21** having a vertical length H**2** smaller than the vertical length H**1** of **139***m***1** in the word lines **1***b* may be provided. A third NMOS gate electrode **121***p***11** having a vertical length H**2** smaller than the vertical length H**1** of the word lines (**139***m***1** in FIG. **1**B) may be provided. A third PMOS gate electrode **121***p***21** having a vertical length H**2** smaller than the vertical length H**1** of the word lines (**139***m***1** in FIG. **1**B) may be provided.

[0149] A second peripheral NMOS transistor N_TRb including a peripheral NMOS gate electrode **139***p***11** having a vertical length smaller than that of the first peripheral NMOS transistor N_TRa in FIG. **1**B may be provided. A second peripheral PMOS transistor P_TRb including a peripheral PMOS gate electrode **139***p***21** having a reduced vertical length smaller than that of the first peripheral PMOS transistor P_TRa in FIG. **1**B may be provided.

[0150] In an example, referring to FIG. **3**B, a peripheral NMOS gate electrode **139***p***12** having a vertical length H**3** greater than the vertical length H**1** of the word lines (**139***m***1** in FIG. **1**B) may be provided. A peripheral PMOS gate electrode **139***p***22** having a vertical length H**3** greater than the vertical length H**1** of the word lines (**139***m***1** in FIG. **1**B) may be provided. A third NMOS gate electrode **121***p***12** having a vertical length H**3** greater than the vertical length H**1** of the word lines (**139***m***1** in FIG. **1**B) may be provided. A third PMOS gate electrode **121***p***22** having a vertical length H**3** greater than the vertical length H**1** of the word lines (**139***m***1** in FIG. **1**B) may be provided.

[0151] A third peripheral NMOS transistor N_TRc including a peripheral NMOS gate electrode **139***p***12** having a vertical length greater than that of the first peripheral NMOS transistor N_TRa in FIG. **1**B may be provided. A third peripheral PMOS transistor P_TRc including a peripheral PMOS gate electrode **139***p***22** having a vertical length greater than that of the first peripheral PMOS transistor P_TRa in FIG. **1**B may be provided.

[0152] In an example, referring to FIG. **3**C, an NMOS gate dielectric layer FOna having a thickness smaller than that of the NMOS gate dielectric layer (FOn in FIG. **1**B) in FIG. **1**B may be provided. A PMOS gate dielectric layer FOpa having a thickness smaller than that of the PMOS gate dielectric layer in FIG. **2**A (FOp in FIG. **1**B) may be provided.

[0153] A fourth peripheral NMOS transistor N_TRd including the NMOS gate dielectric layer FOna having a thickness smaller than that of the first peripheral NMOS transistor N_TRa in FIG. **1**B may be provided. A fourth peripheral PMOS transistor P_TRd including the PMOS gate dielectric layer FOpa having a thickness smaller than that of the first peripheral PMOS transistor P_TRa in FIG. **1**B may be provided.

[0154] In an example, referring to FIG. **3**D, an NMOS gate dielectric layer FOnb having a thickness greater than that of the NMOS gate dielectric layer (FOn in FIG. **1**B) in FIG. **1**B may be provided. A PMOS gate dielectric layer FOpb having a thickness greater than that of the PMOS gate dielectric layer (FOp in FIG. **1**B) in FIG. **1**B may be provided.
[0155] A fifth peripheral NMOS transistor N_TRe including the NMOS gate dielectric layer FOnb a thickness greater than that of the first peripheral NMOS transistor N_TRa in FIG. **1**B may be provided. A fifth peripheral PMOS transistor P_TRe including the PMOS gate dielectric layer FOpb having a thickness greater than the first peripheral PMOS transistor P_TRa in FIG. **1**B may be provided.
[0156] In an example, referring to FIG. **3**E, an additional NMOS source/drain region **188***p***1** may be disposed on the upper NMOS source/drain region **187***p***1** as in FIG. **1**B. An additional PMOS source/drain region **188***p***2** may be disposed on the upper PMOS source/drain region **187***p***2** as illustrated in FIG. **1**B.
[0157] The additional NMOS source/drain region **188***p***1** may have an upper surface in contact with a lower surface of the peripheral NMOS contact structure **154***p***1**. The additional PMOS source/drain region **188***p***2** may have an upper surface in contact with a lower surface of the peripheral PMOS contact structure **154***p***2**.
[0158] The additional NMOS source/drain region **188***p***1** may be formed of silicon having N-type conductivity, and the additional PMOS source/drain region **188***p***2** may be formed of silicon having P-type conductivity. The additional NMOS source/drain region **188***p***1** may be formed of polysilicon having N-type conductivity or epitaxial silicon having N-type conductivity. The additional PMOS source/drain region **188***p***2** may be formed of polysilicon having P-type conductivity or epitaxial silicon having P-type conductivity.
[0159] A sixth peripheral NMOS transistor N_TRf including the additional NMOS source/drain region **188***p***1** may be provided rather than the first peripheral NMOS transistor N_TRa as illustrated in FIG. **1**B. A sixth peripheral PMOS transistor P_TRf including the additional PMOS source/drain region **188***p***2** may be provided rather than the first peripheral PMOS transistor P_TRa as illustrated in FIG. **1**B.
[0160] In an example, referring to FIG. **3**F, the additional NMOS source/drain region **188***p***1** as in FIG. **3**E may be replaced with a merged NMOS source/drain region **188***p***1***a* connected to the NMOS vertical active patterns **109***p***1** adjacent to each other. The additional PMOS source/drain region **188***p***2** as illustrated in FIG. **3**E may be replaced with a merged PMOS source/drain region **188***p***2***a* connected to the PMOS vertical active patterns **109***p***2** adjacent to each other.
[0161] The merged NMOS source/drain region **188***p***1***a* may vertically overlap the third NMOS gate electrode **121***p***1**. The merged PMOS source/drain region **188***p***2***a* may vertically overlap the third PMOS gate electrode **121***p***2**.
[0162] The merged NMOS source/drain region **188***p***1***a* may be formed of silicon having N-type conductivity, and the merged PMOS source/drain region **188***p***2***a* may be formed of silicon having P-type conductivity. The merged NMOS source/drain region **188***p***1***a* may be formed of polysilicon having N-type conductivity or epitaxial silicon having N-type conductivity. The merged PMOS source/drain region **188***p***2***a* may be formed of polysilicon having P-type conductivity or epitaxial silicon having P-type conductivity.
[0163] A seventh peripheral NMOS transistor N_TRg further including the merged NMOS source/drain region **188***p***1***a* rather than the first peripheral NMOS transistor N_TRa as illustrated in FIG. **1**B may be provided. A seventh peripheral PMOS transistor P_TRg further including the merged PMOS source/drain region **188***p***2***a* rather than the first peripheral PMOS transistor P_TRa as illustrated in FIG. **1**B may be provided.
[0164] In FIGS. **1**A to **3**F described above, the first peripheral circuit region PR_U may include at least one of the first to seventh peripheral NMOS transistors N_TRa, N_TRb, N_TRc, N_TRd, N_TRe, N_TRf, and N_TRg, and may include at least one of the first to seventh peripheral PMOS

transistors P_TRa, P_TRb, P_TRc, P_TRd, P_TRe, P_TRf, and P_TRg. Accordingly, the first peripheral circuit region PR_U may form various peripheral circuits using the first to seventh peripheral NMOS transistors N_TRa, N_TRb, N_TRc, N_TRd, N_TRe, N_TRf, and N_TRg and the first to seventh peripheral PMOS transistors P_TRa, P_TRb, P_TRc, P_TRd, P_TRe, P_TRf, and P_TRg.

[0165] In the description below, example embodiments of various circuits having cross-sectional structures using the first to seventh peripheral NMOS transistors N_TRa, N_TRb, N_TRc, N_TRd, N_TRe, N_TRf, N_TRg, the first to seventh peripheral PMOS transistors P_TRa, P_TRb, P_TRc, P_TRd, P_TRe, P_TRf, and P_TRg, the third peripheral NMOS gate electrode **121***p***1** and the third peripheral PMOS gate electrode **121***p***2** in the first peripheral circuit region PR_U will be described with reference to FIGS. **4**A to **4**D. FIGS. **4**A to **4**D are enlarged diagrams illustrating various examples included in a circuit in a semiconductor device according to an example embodiment.

[0166] In an example, referring to FIG. **4**A, the first NMOS vertical active pattern nACT**1** and the second NMOS vertical active pattern nACT**2** may be disposed adjacent to each other. The first NMOS vertical active pattern nACT**1** may include a first lower NMOS source/drain region nSD_L**1**, a first upper NMOS source/drain region nSD_U**1** on the first lower NMOS source/drain region nSD_L**1**, and a first NMOS channel region nCH_**1** between the first lower NMOS source/drain region nSD_L**1** and the first upper NMOS source/drain region nSD_U**1**. The second NMOS vertical active pattern nACT**2** may include a second lower NMOS source/drain region nSD_L**2**, a second upper NMOS source/drain region nSD_U**2** on the second lower NMOS source/drain region nSD_L**2**, and a second NMOS channel region nCH_**2** between the second lower NMOS source/drain region nSD_L**2** and the second upper NMOS source/drain region nSD_U**2**.

[0167] The first NMOS vertical active pattern nACT**1** and the second NMOS vertical active pattern nACT**2** may be substantially the same as the NMOS vertical active patterns **109***p***1** as illustrated in FIG. **1**B.

[0168] The NMOS gate electrode nFG may be disposed between the first NMOS channel region nCH_**1** of the first NMOS vertical active pattern nACT**1** and the second NMOS channel region nCH_**2** of the second NMOS vertical active pattern nACT**2**.

[0169] The first NMOS gate dielectric layer nFOx**1** may be disposed between the first NMOS vertical active pattern nACT**1** and the NMOS gate electrode nFG, and the second NMOS gate dielectric layer nFOx**2** may be disposed between the second NMOS vertical active pattern nACT**2** and the NMOS gate electrode nFG.

[0170] The first NMOS back gate electrode nBG**1** and the second NMOS back gate electrode nBG**2** may be disposed opposite each other. The first NMOS channel region nCH_**1**, the NMOS gate electrode nFG, and the second NMOS channel region nCH_**2** may be disposed between the first NMOS back gate electrode nBG**1** and the second NMOS back gate electrode nBG**2**.

[0171] The first NMOS back gate dielectric layer nBOx**1** may be disposed between the first NMOS back gate electrode nBG**1** and the first NMOS vertical active pattern nACT**1**, and the second NMOS back gate dielectric layer nBOx**2** may be disposed between the second NMOS back gate electrode nBG**2** and the second NMOS vertical active pattern nACT**2**.

[0172] In an example, referring to FIG. **4**B, the first PMOS vertical active pattern pACT**1** and the second PMOS vertical active pattern pACT**2** may be disposed adjacent to each other. The first PMOS vertical active pattern pACT**1** may include a first lower PMOS source/drain region PSD_L**1**, a first upper PMOS source/drain region PSD_U**1** on the first lower PMOS source/drain region PSD_L**1**, and a first PMOS channel region pCH_**1** between the first lower PMOS source/drain region PSD_L**1** and the first upper PMOS source/drain region PSD_U**1**. The second PMOS vertical active pattern pACT**2** may include a second lower PMOS source/drain region PSD_L**2**, a second upper PMOS source/drain region PSD_U**2** on the second lower PMOS source/drain region PSD_L**2**, and a second PMOS channel region pCH_**2** between the second

lower PMOS source/drain region PSD_L**2** and the second upper PMOS source/drain region PSD_U**2**.
[0173] The first PMOS vertical active pattern pACT**1** and the second PMOS vertical active pattern pACT**2** may be substantially the same as the PMOS vertical active patterns **109***p***2** as illustrated in FIG. **1**B.
[0174] The PMOS gate electrode pFG may be disposed between the first PMOS channel region pCH_**1** of the first PMOS vertical active pattern pACT**1** and the second PMOS channel region pCH_**2** of the second PMOS vertical active pattern pACT**2**.
[0175] The first PMOS gate dielectric layer pFOx**1** may be disposed between the first PMOS vertical active pattern pACT**1** and the PMOS gate electrode pFG, and the second PMOS gate dielectric layer pFOx**2** may be disposed between the second PMOS vertical active pattern pACT**2** and the PMOS gate electrode pFG.
[0176] The first PMOS back gate electrode pBG**1** and the second PMOS back gate electrode pBG**2** may be disposed opposite each other. The first PMOS channel region pCH_**1**, the PMOS gate electrode pFG, and the second PMOS channel region pCH_**2** may be disposed between the first PMOS back gate electrode pBG**1** and the second PMOS back gate electrode pBG**2**.
[0177] The first PMOS back gate dielectric layer pBOx**1** may be disposed between the first PMOS back gate electrode pBG**1** and the first PMOS vertical active pattern pACT**1**, and the second PMOS back gate dielectric layer pBOx**2** may be disposed between the second PMOS back gate electrode pBG**2** and the second PMOS vertical active pattern pACT**2**.
[0178] In an example, referring to FIG. **4**C, an NMOS vertical active pattern nACT and a PMOS vertical active pattern pACT may be disposed adjacent to each other. The NMOS vertical active pattern nACT may include a lower NMOS source/drain region nSD_L, an upper NMOS source/drain region nSD_U on the lower NMOS source/drain region nSD_L, and an NMOS channel region nCH between the lower NMOS source/drain region nSD_L and the upper NMOS source/drain region nSD_U. The PMOS vertical active pattern pACT may include a lower PMOS source/drain region PSD_L, an upper PMOS source/drain region pSD_U on the lower PMOS source/drain region PSD_L, and a PMOS channel region pCH between the lower PMOS source/drain region pSD_L and the upper PMOS source/drain region pSD_U.
[0179] A shared back gate electrode BG may be disposed between the NMOS channel region nCH of the NMOS vertical active pattern nACT and the PMOS channel region pCH of the PMOS vertical active pattern pACT.
[0180] The NMOS back gate dielectric layer nBOx may be disposed between the NMOS vertical active pattern nACT and the shared back gate electrode BG, and the PMOS back gate dielectric layer pBOX may be disposed between the PMOS vertical active pattern pACT and the shared back gate electrode BG.
[0181] An NMOS gate electrode nFG and a PMOS gate electrode pFG may be disposed opposite each other. The NMOS channel region nCH, the shared back gate electrode BG, and the PMOS channel region pCH may be disposed between the NMOS gate electrode nFG and the PMOS gate electrode pFG.
[0182] The NMOS gate dielectric layer nFOx may be disposed between the NMOS gate electrode nFG and the NMOS vertical active pattern nACT, and the PMOS gate dielectric layer pFOx may be disposed between the PMOS gate electrode pFG and the PMOS vertical active pattern pACT.
[0183] In an example, referring to FIG. **4**D, the NMOS vertical active pattern nACT and the PMOS vertical active pattern pACT may be disposed adjacent to each other as in FIG. **4**C.
[0184] The shared gate electrode FG may be disposed between the NMOS channel region nCH of the NMOS vertical active pattern nACT and the PMOS channel region pCH of the PMOS vertical active pattern pACT.
[0185] The NMOS gate dielectric layer nFOx may be disposed between the NMOS vertical active pattern nACT and the shared gate electrode FG, and the PMOS gate dielectric layer pFOx may be

disposed between the PMOS vertical active pattern pACT and the shared gate electrode FG.
[0186] An NMOS back gate electrode nBG and a PMOS back gate electrode pBG may be disposed opposite each other. The NMOS channel region nCH, the shared gate electrode FG, and the PMOS channel region pCH may be disposed between the NMOS back gate electrode nBG and the PMOS back gate electrode pBG.
[0187] The NMOS back gate dielectric layer nBOx may be disposed between the NMOS back gate electrode nBG and the NMOS vertical active pattern nACT, and the PMOS back gate dielectric layer pBOx may be disposed between the PMOS back gate electrode pBG and the PMOS vertical active pattern pACT.
[0188] In the description below, example embodiments of planar shapes of various circuits which may be formed using the first to seventh peripheral NMOS transistors N_TRa, N_TRb, N_TRc, N_TRd, N_TRe, N_TRf, and N_TRg, the first to seventh peripheral PMOS transistors P_TRa, P_TRb, P_TRc, P_TRd, P_TRe, P_TRf, and P_TRg, the third peripheral NMOS gate electrode **121***p***1** and the third peripheral PMOS gate electrode **121***p***2** in the first peripheral circuit region PR_U will be described with reference to FIGS. **5**A and **5**B. FIGS. **5**A and **5**B are plan diagrams illustrating various examples included in a circuit in a semiconductor device according to an example embodiment.
[0189] In an example, referring to FIG. **5**A, a first NMOS vertical active pattern nACT**1** and a second NMOS vertical active pattern nACT**2** may be disposed adjacent to each other in the second horizontal direction Y.
[0190] A plurality of the first NMOS vertical active pattern nACT**1** may be arranged in order in the first horizontal direction X. A plurality of the second NMOS vertical active pattern nACT**2** may be arranged in order in the first horizontal direction.
[0191] The first NMOS vertical active pattern nACT**1** and the second NMOS vertical active pattern nACT**2** may be substantially the same as the NMOS vertical active patterns **109***p***1** as in FIG. **1**B.
[0192] The back gate electrode nBG may be disposed between the first NMOS vertical active pattern nACT**1** and the second NMOS vertical active pattern nACT**2**. The back gate electrode nBG may be substantially the same as the third peripheral NMOS gate electrode **121***p***1** as in FIG. **1**B.
[0193] The first NMOS gate electrode nFG**1** and the second NMOS gate electrode nFG**2** may be disposed opposite each other. The first NMOS vertical active pattern nACT**1**, the second NMOS vertical active pattern nACT**2**, and the back gate electrode nBG may be disposed between the first NMOS gate electrode nFG**1** and the second NMOS gate electrode nFG**2**.
[0194] The routing wiring structure nCNT may be electrically connected to each of the back gate electrode nBG, the first NMOS gate electrode nFG**1** and the second NMOS gate electrode nFG**2**. The routing wiring structure nCNT may operate the back gate electrode nBG, the first NMOS gate electrode nFG**1** and the second NMOS gate electrode nFG**2** as a gate electrode. Accordingly, an NMOS transistor using the back gate electrode nBG, the first NMOS gate electrode nFG**1** and the second NMOS gate electrode nFG**2** as a gate electrode may be provided.
[0195] In an example, referring to FIG. **5**B, the first PMOS vertical active pattern pACT**1** and the second PMOS vertical active pattern pACT**2** may be disposed adjacent to each other in the second horizontal direction Y.
[0196] A plurality of the first PMOS vertical active pattern pACT**1** may be arranged in order in the first horizontal direction X. A plurality of the second PMOS vertical active pattern pACT**2** may be arranged in order in the first horizontal direction X.
[0197] The first PMOS vertical active pattern pACT**1** and the second PMOS vertical active pattern pACT**2** may be substantially the same as the PMOS vertical active patterns **109***p***1** as in FIG. **1**B.
[0198] The back gate electrode pBG may be disposed between the first PMOS vertical active pattern pACT**1** and the second PMOS vertical active pattern pACT**2**. The back gate electrode pBG may be substantially the same as the third peripheral PMOS gate electrode **121***p***2** as in FIG. **1**B.
[0199] The first PMOS gate electrode pFG**1** and the second PMOS gate electrode pFG**2** may be

disposed opposite each other. The first PMOS vertical active pattern pACT**1**, the second PMOS vertical active pattern pACT**2**, and the back gate electrode pBG may be disposed between the first PMOS gate electrode pFG**1** and the second PMOS gate electrode pFG**2**.

[0200] The routing wiring structure pCNT may be electrically connected to each of the back gate electrode pBG, the first PMOS gate electrode pFG**1** and the second PMOS gate electrode pFG**2**. The routing wiring structure pCNT may operate the back gate electrode pBG, the first PMOS gate electrode pFG**1** and the second PMOS gate electrode pFG**2** as a gate electrode. Accordingly, a PMOS transistor using the back gate electrode pBG, the first PMOS gate electrode pFG**1** and the second PMOS gate electrode pFG**2** as a gate electrode may be provided.

[0201] In the description below, an example embodiment of an inverter formed using the first to seventh peripheral NMOS transistors N_TRa, N_TRb, N_TRc, N_TRd, N_TRe, N_TRf, and N_TRg, the first to seventh peripheral PMOS transistors P_TRa, P_TRb, P_TRc, P_TRd, P_TRe, P_TRf, and P_TRg, the third peripheral NMOS gate electrode **121***p***1** and the third peripheral PMOS gate electrode **121***p***2** in the first peripheral circuit region PR_U will be described with reference to FIGS. **6**A and **6**B. FIG. **6**A is a plan diagram illustrating an example embodiment of an inverter in a semiconductor device according to an example embodiment, and FIG. **6**B is a cross-sectional diagram illustrating a region corresponding to region “B” in FIG. **1**B, taken along line I-I′ in FIG. **6**A.

[0202] In an example, referring to FIGS. **6**A and **6**B, the inverter INT may be configured using at least one of the first to seventh peripheral NMOS transistors N_TRa, N_TRb, N_TRc, N_TRd, N_TRe, N_TRf, and N_TRg, and at least one of the first to seventh peripheral PMOS transistors P_TRa, P_TRb, P_TRc, P_TRd, P_TRe, P_TRf, and P_TRg, the third peripheral NMOS gate electrode **121***p***1** and the third peripheral PMOS gate electrode **121***p***2** in the first peripheral circuit region PR_U described above. For example, the inverter INT may include the first peripheral NMOS transistors N_TRa and the first peripheral PMOS transistors P_TRa, and the third NMOS peripheral gate electrode **121***p***1** and the third peripheral PMOS gate electrode **121***p***2** described with reference to FIGS. **1**A and **1**B.

[0203] The inverter INT may include the first and second NMOS vertical active patterns **109***p***1***a* and **109***p***1***b* adjacent to each other in the second horizontal direction Y, and the first and second PMOS vertical active patterns **109***p***2***a* and **109***p***2***b* adjacent to each other in the second horizontal direction Y.

[0204] A plurality of the first NMOS vertical active pattern **109***p***1***a* may be arranged and spaced apart from each other in the first horizontal direction X. A plurality of the second NMOS vertical active pattern **109***p***1***b* may be arranged and spaced apart from each other in the first horizontal direction X.

[0205] A plurality of the first PMOS vertical active pattern **109***p***2***a* may be arranged and spaced apart from each other in the first horizontal direction X. A plurality of the second PMOS vertical active pattern **109***p***2***b* may be arranged and spaced apart from each other in the first horizontal direction X.

[0206] The inverter INT may include the first lower peripheral wiring **190***p***1** and the second lower peripheral wiring **190***p***2** described with reference to FIG. **1**B. The first lower peripheral wiring **190***p***1** described with reference to FIG. **1**B may be connected to lower surfaces of the first NMOS vertical active patterns **109***p***1***a*, and the second lower peripheral wiring **190***p***2** described with reference to FIG. **1**B may be connected to lower surfaces of the first PMOS vertical active patterns **109***p***2***a*.

[0207] The inverter INT may further include a gate contact structure G_C electrically connecting the first NMOS gate electrode **139***p***1***a*, the second NMOS gate electrode **139***p***1***b*, the first PMOS gate electrode **139***p***2***a* and the second PMOS gate electrode **139***p***2***b* to each other, as described with reference to FIG. **1**B.

[0208] The inverter INT may further include an NMOS back gate contact structure N_BC

electrically connected to the third NMOS gate electrode **121***p***1** and a PMOS back gate contact structure P_BC electrically connected to the third PMOS gate electrode **121***p***2**.
[0209] The inverter INT may include the NMOS contact plugs **154***p***1** connected to the upper surfaces of the first NMOS vertical active patterns **109***p***1***a* and the PMOS contact plugs **154***p***2** connected to the upper surfaces of the first PMOS vertical active patterns **109***p***2***a* as described with reference to FIG. **1**B.
[0210] The inverter INT may include a common upper wiring **160***pba* disposed on the NMOS contact plugs **154***p***1** and the PMOS contact plugs **154***p***2** and electrically connected to the NMOS contact plugs **154***p***1** and the PMOS contact plugs **154***p***2**. The common upper wiring **160***pba* may be electrically connected to the upper NMOS source/drain regions **187***p***1** of the first NMOS vertical active patterns **109***p***1***a* and the upper PMOS source/drain regions **187***p***2** of the first PMOS vertical active patterns **109***p***2***a* through the NMOS contact plugs **154***p***1** and the PMOS contact plugs **154***p***2**.
[0211] Using the first peripheral NMOS transistors N_TRa and the first peripheral PMOS transistors P_TRa, by applying a negative bias to the third NMOS gate electrode **121***m***1** and a positive bias to the third PMOS gate electrode **121***m***2**, performance of the inverter INT may be improved during operation of the inverter INT. The third NMOS gate electrode **121***m***1** may be configured as an NMOS back gate electrode, and the third PMOS gate electrode **121***m***2** may be configured as a PMOS back gate electrode.
[0212] In the description below, example embodiments of the second insulating structure INS, the cell gate dielectric layers FOc, the cell back gate dielectric layers BOc, the NMOS gate dielectric layers GOn, the PMOS gate dielectric layers GOp, the third NMOS gate dielectric layers BOn and the third PMOS gate dielectric layers Bop in FIG. **1**B will be described with reference to FIG. **7**. FIG. **7** is an enlarged cross-sectional diagram illustrating regions corresponding to regions “A” and “B” in FIGS. **1**A and **1**B.
[0213] In an example, referring to FIG. **7**, the second insulating structure INS may include upper gate capping insulating layers **145**, upper back gate capping insulating layers **189**, a buffer insulating layer **133**, and gap-fill insulating layers **142**.
[0214] The upper gate capping insulating layers **145** may be disposed on upper surfaces of the word lines **139***m***1** and the peripheral gate electrodes **139***p***1** and **139***p***2** described with reference to FIG. **1**B. The upper back gate capping insulating layers **189** may be disposed on upper surfaces of the cell back gate electrodes **121***m***1** and the third peripheral gate electrodes **121***p***1** and **121***p***2** described with reference to FIG. **1**B. The buffer insulating layer **133** may be disposed on upper surfaces of the bit lines **190***m***1** and **190***m***2**, the lower peripheral wirings **190***p***1** and **190***p***2** and the first insulating structure **198**. The gap-fill insulating layers **142** may be disposed between the word lines **139***m***1** without the cell vertical active pattern **109***m***1** therebetween among the word lines **139***m***1** adjacent to each other, and may be disposed between the peripheral gate electrodes **139***p***1** and **139***p***2** without the peripheral vertical active pattern **109***p***1** and **109***p***2** therebetween among the peripheral gate electrodes **139***p***1** and **139***p***2** adjacent to each other.
[0215] The cell gate dielectric layers FOc described with reference to FIG. **1**B may extend from a portion disposed between the cell vertical active patterns **109***m***1** and the word lines **139***m***1** to regions between the cell vertical active patterns **109***m***1** and the upper gate capping insulating layers **145**, between the lower surfaces of the word lines **139***m***1** and the buffer insulating layer **133**, and between the lower surfaces of the gap-fill insulating layers **142** and the buffer insulating layer **133**.
[0216] The cell back gate dielectric layers BOc described with reference to FIG. **1**B may extend from a portion disposed between the cell vertical active patterns **109***m***1** and the cell back gate electrodes **121***m***1** to regions between the cell vertical active patterns **109***m***1** and the upper back gate capping insulating layers **189**, and between the lower surfaces of the cell back gate electrodes **121***m***1** and the buffer insulating layer **133**.
[0217] The NMOS and PMOS gate dielectric layers GOn and GOp described with reference to

FIG. **1**B may extend from a portion disposed between the peripheral vertical active patterns **109***p***1** and **109***p***2** and the peripheral gate electrodes **139***p***1** and **139***p***2** to regions between the peripheral vertical active patterns **109***p***1** and **109***p***2** and the upper gate capping insulating layers **145**, between the lower surfaces of the peripheral vertical active patterns **109***p***1** and **109***p***2** and the buffer insulating layer **133**, and between the lower surfaces of the gap-fill insulating layers **142** and the buffer insulating layer **133**.

[0218] The third gate dielectric layers BOn and BOp described with reference to FIG. **1**B may extend from a portion disposed between the peripheral vertical active patterns **109***p***1** and **109***p***2** and the third peripheral gate electrodes **121***p***1** and **121***p***2** to regions between the peripheral vertical active patterns **109***p***1** and **109***p***2** and the upper back gate capping insulating layers **189**, and the third peripheral gate electrodes **121***p***1**, and between the lower surfaces of **121***p***2** and the buffer insulating layer **133**.

[0219] In the description below, example embodiments of planar shapes of various circuits configured using the first to seventh peripheral NMOS transistors N_TRa, N_TRb, N_TRc, N_TRd, N_TRe, N_TRf, and N_TRg and the first to seventh peripheral PMOS transistors P_TRa, P_TRb, P_TRc, P_TRd, P_TRe, P_TRf, and P_TRg and, the third peripheral NMOS gate electrode **121***p***1** and the third peripheral PMOS gate electrode **121***p***2** in the first peripheral circuit region PR_U will be described with reference to FIGS. **8**A to **8**F. FIGS. **8**A to **8**F are plan diagrams illustrating various examples of configuring a circuit in a semiconductor device according to an example embodiment.

[0220] In an example, referring to FIG. **8**A, peripheral vertical active patterns ACT may be arranged in the first horizontal direction X and the second horizontal direction Y, perpendicular to each other. Each of the peripheral vertical active patterns ACT may having a bar shape extending in the first horizontal direction X.

[0221] The peripheral vertical active patterns ACT may be the NMOS vertical active pattern **109***p***1** described with reference to FIGS. **1**A and **1**B or the PMOS vertical active pattern **109***p***2** described with reference to FIGS. **1**A and **1**B.

[0222] In the description below, the example in which the peripheral vertical active patterns ACT is the NMOS vertical active pattern **109***p***1** described in FIGS. **1**A and **1**B will be described.

[0223] The peripheral gate electrodes GE may correspond to the NMOS gate electrodes **139***p***1** described with reference to FIGS. **1**A and **1**B. The third peripheral gate electrodes BG, which may correspond to the back gate electrode, may correspond to the third peripheral gate electrode **121***p***1** described with reference to FIGS. **1**A and **1**B, for example.

[0224] Each of the third peripheral gate electrodes BG may extend in the first horizontal direction X. The third peripheral gate electrodes BG may be parallel to each other.

[0225] A pair of peripheral gate electrodes GE among the peripheral gate electrodes GE may be disposed between a pair of the third peripheral gate electrodes BG adjacent to each other.

[0226] Lower peripheral wirings **190**_A may correspond to the first lower peripheral wirings **190***p***1** described with reference to FIGS. **1**A and **1**B. Each of the lower peripheral wirings **190**_A may have a line shape extending in the second horizontal direction Y. The lower peripheral wirings **190**_A may be parallel to each other.

[0227] Peripheral contact structures **154**_A may correspond to the first peripheral contact structures **154***p***1** described with reference to FIGS. **1**A and **1**B. The peripheral contact structures **154**_A may be connected to the peripheral vertical active patterns ACT, respectively. For example, the peripheral contact structure **154**_A may be connected to the peripheral vertical active pattern ACT.

[0228] In another example, referring to FIG. **8**B, the peripheral contact structures **154**_A described with reference to FIG. **8**A may be modified to peripheral contact structures **154**_B as in FIG. **8**B. For example, the peripheral contact structures **154**_B may be connected to a plurality of the peripheral vertical active patterns ACT, respectively. In other words, the peripheral contact structure **154**_B may be connected to the plurality of peripheral vertical active patterns ACT. The

peripheral contact structures **154**_B may be spaced apart from each other. Each of the peripheral contact structures **154**_B may have a circular shape or a quadrangular shape.
[0229] In another example, referring to FIG. **8**C, the peripheral contact structures **154**_A described with reference to FIG. **8**A may be modified to peripheral contact structures **154**_C as in FIG. **8**C. For example, each of the peripheral contact structures **154**_C may have a line shape extending in the second horizontal direction Y, and may be connected to the peripheral vertical active patterns ACT. The peripheral contact structures **154**_C may vertically overlap the lower peripheral wirings **190**_A.
[0230] In another example, referring to FIG. **8**D, the peripheral contact structures **154**_A described with reference to FIG. **8**A may be modified to peripheral contact structures **154**_D as in FIG. **8**D. For example, each of the peripheral contact structures **154**_D may have a line shape extending in the first horizontal direction X, and may be connected to the peripheral vertical active patterns ACT. The peripheral contact structures **154**_D may have a line shape extending in a direction intersecting the lower peripheral wirings **190**_A.
[0231] In another example, referring to FIG. **8**E, the lower peripheral wirings **190**_A in FIGS. **8**A to **8**D may be modified to lower peripheral wirings **190**_B as in FIG. **8**E.
[0232] The lower peripheral wiring **190**_B may be configured as a conductive plate connected to the entire peripheral vertical active patterns ACT.
[0233] In another example, referring to FIG. **8**F, the lower peripheral wirings **190**_A in FIGS. **8**A to **8**D may be modified to lower peripheral wirings **190**_C as in FIG. **8**F.
[0234] Each of the lower peripheral wirings **190**_C may extend in the first horizontal direction X and may be connected to the peripheral vertical active patterns ACT. The lower peripheral wirings **190**_C may be parallel to each other.
[0235] In the description below, an example of the upper peripheral wirings **160***pa* and **160***pb* of the first peripheral circuit region PR_U described with reference to FIGS. **1**A and **1**B will be described with reference to FIG. **9**A. FIG. **9**A may illustrate an example of the upper peripheral wirings **160***pa* and **160***pb* of the first peripheral circuit region PR_U in the cross-sectional structure as in FIG. **1**A.
[0236] Referring to FIG. **9**A, the upper peripheral wirings **160***pa* and **160***pb* described with reference to FIGS. **1**A and **1**B may be replaced with peripheral pad patterns **160***p* disposed on the same level as a level of the cell pad patterns **160***m***1**. Upper peripheral wirings **169***pa* and **169***pb* having substantially the same shape as that of the upper peripheral wirings **160***pa* and **160***pb* described with reference to FIGS. **1**A and **1**B may be disposed on the peripheral pad patterns **160***p*. A portion of the lower region of the second electrode **169** may be disposed on the same level as a level of the upper peripheral wirings **169***pa* and **169***pb*. For example, lower surfaces of the upper peripheral wirings **169***pa*, **169***pb* may be disposed on a level lower than a level of the second electrode **169**, and upper surfaces of the upper peripheral wirings **169***pa*, **169***pb* may be disposed on a level higher than a level of the lower surface of the second electrode **169** and lower than the upper surface of the second electrode **169**. The upper peripheral wirings **169***pa* and **169***pb* may be disposed on the same level as a level of a portion of the second electrode **169**. The upper peripheral wirings **169***pa* and **169***pb* may be formed of the same conductive material as that of the second electrode **169**.
[0237] In the description below, an example of a semiconductor device according to an example embodiment will be described with reference to FIG. **9**B. FIG. **9**B is a cross-sectional diagram illustrating an example of a semiconductor device according to an example embodiment.
[0238] Referring to FIG. **9**B, the structure including the first and second structures CH**1** and CH**2** in FIG. **1**A may be disposed upside down. For example, the second structure CH**2** may be disposed on the first structure CH**1**.
[0239] The first structure CHI may further include a routing structure **179***m* electrically connecting the cell upper wirings **184***m* to the first and second bonding pads **35** and **196** between the cell upper

wirings **184***m* and the second structure CH**2**. In the first structure CH**1**, the bit lines **190***m***1** and **190***m***2** may be closer to the second structure CH**2** than the data storage structures DS**1** and DS**2**.
[0240] The second structure CH**2** may further include a backside insulating layer **50** on the substrate **5**, a through-electrode **60** penetrating the backside insulating layer **50** and the substrate **5** and electrically connected to the second routing wiring structure **30**, an insulating spacer **55** on a side surface of the through-electrode **60**, and a conductive pad **70** connected to the through-electrode **60** on the backside insulating layer **50**.
[0241] In the description below, an example of a semiconductor device according to an example embodiment will be described with reference to FIG. **10**. FIG. **10** is a cross-sectional diagram illustrating an example of a semiconductor device according to an example embodiment.
[0242] Referring to FIG. **10**, the first structure CHI having the first and second memory regions MR**1** and MR**2** and the first peripheral circuit region PR_U described with reference to FIGS. **1**A and **1**B may be modified to first structure CH**1***a* having first and second memory regions MR**1** and MR**2** and first peripheral circuit region PR_U as in FIG. **10**. The second structure CH**2** having the first and second core circuit regions CRI and CR**2** and the second peripheral circuit region PR_L described in FIGS. **1**A and **1**B may be modified to a second structure CH**2***a* having first and second core circuit regions CR**1** and CR**2** and second peripheral circuit region PR_L as in FIG. **10**. The second structure CH**2***a* may be disposed on the first structure CH**1***a* as illustrated in FIG. **10**.
[0243] The first structure CH**1***a* may be substantially the same as the first structure (CH**1** in FIG. **1**A) in FIG. **1**A in which the first bonding pads (**196** in FIG. **1**A) is not provided. Accordingly, the first insulating structure **198** as in FIG. **1**A may be bonded to the second structure CH**2***a*.
[0244] The second structure CH**2***a* may include a substrate **205** and a device isolation region **210** limiting active regions **210***a* on the substrate **205**. The substrate **205** may be configured as a semiconductor substrate.
[0245] The first structure CH**1** may further include a routing structure **179***m* electrically connecting the cell upper wirings **184***m* to the second routing wiring structure **192** between the cell upper wirings **184***m* and the second structure CH**2**. In the first structure CH**1**, the bit lines **190***m***1** and **190***m***2** may be closer to the second structure CH**2** than the data storage structures DS**1** and DS**2**.
[0246] The second structure CH**2***a* may include a lower peripheral circuit transistors **225***a*, **225***b*, **225***c* disposed on the substrate **205**, a second routing wiring structure **230** electrically connected to the lower peripheral circuit transistors **225***a*, **225***b*, and **225***c* on the lower peripheral circuit transistors **225***a*, **225***b*, and **225***c*, and a lower insulating structure **240** covering the lower peripheral circuit transistors **225***a*, **225***b*, **225***c* and the second routing wiring structure **230** on the substrate **205**.
[0247] The lower peripheral circuit transistors **225***a*, **225***b*, and **225***c* may include peripheral gate structures **215***a*, **215***b* and peripheral source/drain regions **220***sd*, respectively. The peripheral gate structures **215***a* and **215***b* may include a peripheral gate dielectric layer **215***a* and a peripheral gate electrode **215***b*.
[0248] The lower peripheral circuit transistors **225***a*, **225***b*, and **225***c* may include a first core circuit transistor **225***a* in the first core circuit region CR**1**, a second core circuit transistor **225***b* in the second core circuit region CR**2**, and a second peripheral circuit transistor **225***c* in the second peripheral circuit region PR_L.
[0249] The first core circuit region CR**1** may include the first core circuit transistor **225***a*. The second core circuit region CR**2** may include the second core circuit transistor **225***b*. The first core circuit transistor **225***a* may vertically overlap the first data storage structure DS**1**, and the second core circuit transistor **225***b* may vertically overlap the second data storage structure DS**2**.
[0250] The second structure CH**2***a* may be disposed below the substrate **205** and may further include a backside insulating layer **250** bonded to the first insulating structure **198**. The lower surface of the backside insulating layer **250** and the upper surface of the first insulating structure **198** may be bonded to form a bonding surface JSa between the first structure CH**1***a* and the second

structure CH**2***a*.

[0251] The second structure CH**2***a* may include through-electrodes **260***a*, **260***b*, **260***c*, and insulating spacers **265** surrounding a side surface of each of the through-electrodes **260***a*, **260***b*, and **260***c*. Through-electrodes **260***a*, **260***b*, **260***c* may penetrate the substrate **5**, extend upwardly, and connect to the second routing wiring structure **230**, and may penetrate the substrate **5** and the insulating layer **250**, extend downwardly and connect to the first routing wiring structure **192** in the first structure CH**1***a*.

[0252] The upper surfaces of the through-electrodes **260***a*, **260***b*, and **260***c* may be connected to the second routing wiring structure **230**, and the lower surfaces of the through-electrodes **260***a*, **260***b*, and **260***c* may be connected to the first routing structure **192**.

[0253] The through-electrodes **260***a*, **260***b*, and **260***c* may include a first through-electrode **260***a* electrically connecting the first core circuit transistor **225***a* to the first bit line **190***m***1**, a second through-electrode **260***b* electrically connecting the second core circuit transistor **225***b* to the second bit line **190***m***2**, and a third through-electrode **260***c* disposed in the first and second peripheral circuit regions PR_L and PR_U.

[0254] The second structure CH**2***a* may further include a conductive pad **285** disposed on the lower insulating structure **240** and electrically connected to the second routing wiring structure **230**.

[0255] In the description below, an example of a semiconductor device according to an example embodiment will be described with reference to FIG. **11**. FIG. **11** is a cross-sectional diagram illustrating an example of a semiconductor device according to an example embodiment.

[0256] Referring to FIG. **11**, the first structure CH**1** having the first and second memory regions MR**1** and MR**2** and the first peripheral circuit region PR_U described with reference to FIGS. **1**A and **1**B may be modified to first structure CH**1***b* having first and second memory regions MR**1** and MR**2** and first peripheral circuit region PR_U as in FIG. **11**. The second structure CH**2** having the first and second core circuit regions CR**1** and CR**2** and the second peripheral circuit region PR_L described in FIGS. **1**A and **1**B may be modified to a second structure CH**2***b* having first and second core circuit regions CR**1** and CR**2** and second peripheral circuit region PR_L as in FIG. **11**.

[0257] The first structure CH**1***b* may not be provided with the first bonding pads (**196** in FIG. **1**A), the first routing wiring structure **192**, the connection contact plugs **194**, the cell upper wirings **184***m* and the third peripheral upper wiring **184***p*, differently from the first structure (CH**1** in FIG. **1**A), and may further include an upper routing wiring structure **384** disposed in the upper insulating structure **181**, and a first bonding pads **335** having an upper surface coplanar with the upper surface of the upper insulating structure **181** and electrically connected to the upper routing wiring structure **384**, differently from the first structure (CH**1** in FIG. **1**A).

[0258] The first structure CH**1***b* may further include connection routing structures **394** connecting the first and second bit lines **190***m***1** and **190***m***2** to the upper routing wiring structure **384**, differently from the first structure (CH**1** in FIG. **1**A) in FIG. **1**A.

[0259] Each of the connection routing structures **394** may include an upper plug **394***c* disposed on the same level as a level of the peripheral contact plug **178***p* and connected to the upper routing wiring structure **384**, an intermediate pad pattern **394***b* disposed on the same level as a level of the cell pad pattern **160***m*, and a lower plug **394***a* disposed between the first and second bit lines **190***m***1** and **190***m***2** and the intermediate pad pattern **394***b*.

[0260] The second structure CH**2***b* may be disposed on the first structure CH**1***b*.

[0261] The second structure CH**2***b* may include a substrate **405** and a device isolation region **410** limiting the active regions **410***a* below the substrate **405**. The substrate **405** may be configured as a semiconductor substrate.

[0262] The second structure CH**2***b* may include lower peripheral circuit transistors **425***a*, **425***b*, **425***c* disposed below the substrate **405**, a second routing wiring structure **430** electrically connected to the lower peripheral circuit transistors **425***a*, **425***b*, and **425***c* below the lower peripheral circuit transistors **425***a*, **425***b*, and **425***c*, a lower insulating structure **440** covering the lower peripheral circuit transistors **425***a*, **425***b*, **425***c*, the second routing wiring structure **430** under the substrate **405**, and first bonding pads **435** having a lower surface coplanar with a lower surface of the lower insulating structure **440**.

[0263] Each of the lower peripheral circuit transistors **425***a*, **425***b*, and **425***c* may include peripheral gate structures **415***a* and **415***b* and peripheral source/drain regions **420***sd*. The peripheral gate structures **415***a* and **415***b* may include a peripheral gate dielectric layer **415***a* and a peripheral gate electrode **415***b*.

[0264] The lower peripheral circuit transistors **425***a*, **425***b*, and **425***c* may include a first core circuit transistor **425***a* in the first core circuit region CR**1**, a second core circuit transistor **425***b* in the second core circuit region CR**2**, and a second peripheral circuit transistor **425***c* in the second peripheral circuit region PR_L.

[0265] The first core circuit region CR**1** may include the first core circuit transistor **425***a*. The second core circuit region CR**2** may include the second core circuit transistor **425***b*. The first core circuit transistor **425***a* may vertically overlap the first data storage structure DS**1**, and the second core circuit transistor **425***b* may vertically overlap the second data storage structure DS**2**.

[0266] A lower surface of the lower insulating structure **440** of the second structure CH**2***b* may be bonded to an upper surface of the upper insulating structure **181** of the first structure CH**1***b*, and lower surfaces of the second bonding pads **435** of the second structure CH**2***b* may be bonded to the upper surface of the upper insulating structure **181** of the first structure CH**1***b*. The second structure CH**2***b* may be bonded to upper surfaces of the first bonding pads **335** of the first structure CH**1***b*. Accordingly, the bonding surface JSc formed by bonding the upper surface of the first structure CH**1***b* to the lower surface of the second structure CH**2***b* may be formed.

[0267] The second structure CH**2***b* may further include a backside insulating layer **450** on the substrate **405**, a through-electrode **460** penetrating the backside insulating layer **450** and the substrate **405** and electrically connected to the second routing wiring structure **430**, an insulating spacer **455** on a side surface of the through-electrode **460**, and a conductive pad **470** connected to the through-electrode **460** on the backside insulating layer **450**.

[0268] In the description below, an example of a semiconductor device according to an example embodiment will be described with reference to FIG. **12**. FIG. **12** is a cross-sectional diagram illustrating an example of a semiconductor device according to an example embodiment.

[0269] Referring to FIG. **12**, the first structure CH**1***b* having the first and second memory regions MR**1** and MR**2** and the first peripheral circuit region PR_U described with reference to FIG. **11** may be modified to first structure CH**1***c* with first and second memory regions MR**1** and MR**2** and first peripheral circuit region PR_U as in FIG. **12**. The second structure CH**2***b* having the first and second core circuit regions CR**1** and CR**2** and the second peripheral circuit region PR_L described in FIG. **11** may be modified to a second structure CH**2***c* having first and second core circuit regions CR**1** and CR**2** and second peripheral circuit region PR_L as in FIG. **12**.

[0270] The first structure CH**1***c* may be substantially the same as the first structure (CH**1***b* in FIG. **11**) in FIG. **11** in which the first bonding pads (**335** in FIG. **11**) is not provided. Accordingly, the upper surface of the upper insulating structure **181**, as illustrated in FIG. **12**, may be bonded to the lower surface of the second structure CH**2***c*.

[0271] The second structure CH**2***c* may include a substrate **505** and a device isolation region **510** limiting active regions **510***a* on the substrate **505**. The substrate **505** may be configured as a semiconductor substrate.

[0272] The second structure CH**2***c* may include a lower peripheral circuit transistors **525***a*, **525***b*, and **525***c* disposed on the substrate **505**, a second routing wiring structure **530** electrically connected to the lower peripheral circuit transistors **525***a*, **525***b*, and **525***c* on the lower peripheral circuit transistors **525***a*, **525***b*, and **525***c*, a lower insulating structure **40** covering the lower peripheral circuit transistors **525***a*, **525***b*, and **525***c* and the second routing wiring structure **530** on the substrate **505**, and the insulating layer **550** disposed below the substrate **505**.

[0273] The lower surface of the insulating layer **550** and the upper surface of the upper insulating structure **181** may be bonded to form a bonding surface JSd between the first structure CH**1***c* and the second structure CH**2***c*.
[0274] The lower peripheral circuit transistors **525***a*, **525***b*, and **525***c* may include peripheral gate structures **515***a*, **515***b*, and peripheral source/drain regions **520***sd*, respectively. The peripheral gate structures **515***a* and **515***b* may include a peripheral gate dielectric layer **515***a* and a peripheral gate electrode **515***b*.
[0275] The lower peripheral circuit transistors **525***a*, **525***b*, and **525***c* may include a first core circuit transistor **525***a* in the first core circuit region CR**1**, a second core circuit transistor **525***b* in the second core circuit region CR**2**, and a second peripheral circuit transistor **525***c* in the second peripheral circuit region PR_L.
[0276] The first core circuit region CR**1** may include the first core circuit transistor **525***a*. The second core circuit region CR**2** may include the second core circuit transistor **525***b*. The first core circuit transistor **525***a* may vertically overlap the first data storage structure DS**1**, and the second core circuit transistor **525***b* may vertically overlap the second data storage structure DS**2**.
[0277] The second structure CH**2***c* may include through-electrodes **560***a*, **560***b*, and **560***c*, and insulating spacers **565** surrounding a side surface of each of the through-electrodes **560***a*, **560***b*, and **560***c*. Through-electrodes **560***a*, **560***b*, and **560***c* may penetrate the substrate **5**, extend upwardly, and connect to the second routing wiring structure **530**, and may penetrate the substrate **505** and the insulating layer **550**, extend downwardly and connect to the first routing wiring structure **384***a* in the first structure CH**1***c*.
[0278] The upper surfaces of the through-electrodes **560***a*, **560***b*, and **560***c* may be connected to the second routing wiring structure **530**, and the lower surfaces of the through-electrodes **560***a*, **560***b*, and **560***c* may be connected to the first routing structure **384***a*.
[0279] The through-electrodes **560***a*, **560***b*, and **560***c* may include a first through-electrode **560***a* electrically connecting the first core circuit transistor **525***a* to the first bit line **190***m***1**, a second through-electrode **560***b* electrically connecting the second core circuit transistor **525***b* to the second bit line **190***m***2**, and a third through-electrode **560***c* disposed in the first and second peripheral circuit regions PR_L and PR_U.
[0280] The second structure CH**2***c* may further include a conductive pad **585** disposed on the lower insulating structure **540** and electrically connected to the second routing wiring structure **530**.
[0281] In the description below, an example of a semiconductor device according to an example embodiment will be described with reference to FIG. **13**. FIG. **13** is a cross-sectional diagram illustrating an example of a semiconductor device according to an example embodiment.
[0282] Referring to FIG. **13**, the example embodiments in FIGS. **1**A to **12** may further include a power capacitor CAP_P disposed in the first peripheral circuit region PR_U and having a structure similar to that of the data storage structures DS**1** and DS**2**, and an electrode connection pattern **660***p* connected to the power capacitor CAP_P. For example, the first structure CHI as in FIG. **1**A may further include the power capacitor CAP_P and the electrode connection pattern **660***p*.
[0283] The power capacitor CAP_P may include a first power electrode **663** disposed on the same level as a level of the first electrode **163**, a second power electrode **669** disposed on the same level as a level of the second electrode **169**, and a dielectric layer **666** between the first power electrodes **663** and the second power electrode **669**.
[0284] The electrode connection pattern **660***p* may be disposed on the same level as a level of the cell pad patterns **161***m***1** and **160***m***2**, and may be connected to the lower surfaces of the first power electrodes **663**.
[0285] In the description below, an example embodiment of a semiconductor device according to an example embodiment will be described with reference to FIG. **14**. FIG. **14** is a perspective diagram illustrating a semiconductor device according to an example embodiment.
[0286] Referring to FIG. **14**, in an example, one of the example embodiments in FIGS. **1**A to **13**

may further include a third structure CH**3** to increase integration density. For example, the semiconductor device **1** described with reference to FIGS. **1**A and **1**B may further include the third structure CH**3** disposed on the first structure CH**1**. Similarly, as illustrated in FIG. **13**, the third structure CH**3** may be disposed on the first structure CH**1** including the power capacitor (CAP_P in FIG. **13**). FIG. **14** illustrates an example in which the semiconductor device **1** in FIG. **1**A further includes the third structure CH**3**, but an example embodiment thereof is not limited thereto. For example, in example embodiments as in FIGS. **11** and **13**, the third structure CH**3** may be disposed below the first structures CH**1***b* and CH**1***c*.
[0287] In one of the example embodiments in FIGS. **1**A to **13**, a plurality of the first memory region MR**1** may be arranged in the first horizontal direction X, a plurality of the second memory region MR**2** may be arranged in the first horizontal direction X, a plurality of the first core circuit region CRI may be arranged in the first horizontal direction X, and a plurality of the second core circuit region CR**2** may be arranged in the first horizontal direction X. The plurality of first memory regions MR**1** and the plurality of first core circuit regions CR**1** described above may overlap in the vertical direction Z, and the plurality of second memory regions MR**2** and the plurality of second core circuit regions CR**2** may overlap in the vertical direction Z.
[0288] The first peripheral circuit region PR_U may be disposed between the plurality of first memory regions MR**1** and the plurality of second memory regions MR**2**. The second peripheral circuit region PR_L may be disposed between the plurality of first core circuit regions CR**1** and the plurality of second core circuit regions CR**2**.
[0289] The third structure CH**3** may include a plurality of third memory regions MR**3** vertically overlapping the plurality of first memory regions MR**1**, and a fourth memory regions MR**4** vertically overlapping the plurality of second memory regions MR**2**.
[0290] The third and fourth memory regions MR**3** and MR**4** may include cell transistors (c_TRa and c_TRb in FIGS. **1**A and **1**B) and the data storage structures (DS**1**, DS**2** in FIG. **1**A) substantially the same as the first and second memory regions MR**1** and MR**2**.
[0291] The third structure CH**3** may include a third peripheral circuit region PR_Ua vertically overlapping the first peripheral circuit region PR_U, but in example embodiments, the third peripheral circuit region PR_Ua may not be provided.
[0292] In the other example embodiment, the third and fourth memory regions MR**3** and MR**4** may be formed on the first and second memory regions MR**1** and MR**2** in the first structure CH**1** without separately forming the third structure CH**3**. For example, the first structure CH**1** may further include the third and fourth memory regions MR**3** and MR**4** disposed on the first and second memory regions MR**1** and MR**2**, and vertically overlapping the first and second memory regions MR**1** and MR**2**.
[0293] In the description below, a method of manufacturing a semiconductor device according to an example embodiment will be described with reference to FIG. **15**. FIG. **15** is a flowchart illustrating a method of manufacturing a semiconductor device according to an example embodiment.
[0294] Referring to FIG. **15**, a first structure including the first peripheral circuit region and memory regions may be formed (S**10**). The first structure may be the first structure of one of the example embodiments in FIGS. **1**A to **14**. For example, the first structure may be the first structure CHI in FIG. **1**A or FIG. **13**, the first peripheral circuit region may be the first peripheral circuit region PR_U in FIG. **1**A or FIG. **13**, and the memory regions may be the first and second memory regions MR**1** and MR**2** in FIG. **1**A or **13**.
[0295] A second structure including a second peripheral circuit region and core circuit regions may be formed (S**20**). The second structure may be the second structure of any of the example embodiments in FIGS. **1**A to **14**. For example, the second structure may be the second structure CH**2** in FIG. **1**A or FIG. **13**, the second peripheral circuit region may be the second peripheral circuit region PR_L in FIG. **1**A or FIG. **13**, and the core circuit regions may be the first and second

core circuit regions CR**1** and CR**2** in FIG. **1**A or FIG. **13**.
[0296] The first structure and the second structure may be bonded to each other (S**30**). Accordingly, the first structure and the second structure bonded to each other may be formed. For example, as illustrated in FIG. **1**A, the first structure CHI and the second structure CH**2** bonded to each other may be formed.
[0297] In the description below, a method of manufacturing a semiconductor device according to an example embodiment will be described with reference to FIGS. **16**A to **16**H, and FIG. **17**. FIG. **15** is a flowchart illustrating a method of manufacturing a semiconductor device according to an example embodiment, and FIGS. **16**A to **16**H, and FIG. **17** are cross-sectional diagrams illustrating a semiconductor device according to an example embodiment.
[0298] Referring to FIG. **16**A, a sacrificial substrate **103**, a buffer insulating layer **106**, a semiconductor layer **109** and an insulating layer **112** may be stacked in order. The semiconductor layer **109** may be formed of a semiconductor material such as single crystal silicon.
[0299] Trenches **115** penetrating the insulating layer **112**, the semiconductor layer **109** and the buffer insulating layer **106** may be formed. The trenches **115** may be formed in the first memory region MR**1**, the first peripheral circuit region PR_U, and the second memory region MR**2**, arranged in order in the second horizontal direction Y.
[0300] Each of the trenches **115** may have a line shape extending in the first horizontal direction X. The semiconductor layers **109** may be spaced apart from each other in the second horizontal direction Y by the trenches **115**.
[0301] A back gate dielectric layer **118** conformally covering internal walls of the trenches **115** may be formed, a back gate conductive layer may be formed on the back gate dielectric layer **118**, preliminary back gate electrodes **121** partially filling the trenches **115** may be formed by partially etching the back gate conductive layer through an etch back process, back gate capping insulating layers **124** filling the other portions of the trenches **115** may be formed on the preliminary back gate electrodes **121**. The back gate capping insulating layers **124** may be formed of an insulating material.
[0302] Referring to FIG. **16**B, the mask pattern **127** may be formed, and vertical active patterns **109***m***1**, **109***p***1**, **109***p***2**, and **109***m***2** may be formed by patterning the semiconductor layers (**109** in FIG. **16**A). In the cross-sectional structure as illustrated in FIG. **16**B, pair of vertical active patterns adjacent to each other among the vertical active patterns **109***m***1**, **109***p***1**, **109***p***2**, and **109***m***2** may be formed on both sides of one of the preliminary back gate electrodes **121**.
[0303] The vertical active patterns **109***m***1**, **109***p***1**, **109***p***2**, and **109***m***2** may include first cell vertical active patterns **109***m***1** in the first memory region MR**1**, second cell vertical active patterns **109***m***2** in the second memory region MR**2**, and NMOS vertical active patterns **109***p***1** and PMOS vertical active patterns **109***p***2** in the first peripheral circuit region PR_U.
[0304] The insulating layer (**112** in FIG. **16**A) may include the insulating patterns **112***a* remaining on the vertical active patterns **109***m***1**, **109***p***1**, **109***p***2**, and **109***m***2**. The insulating patterns **112***a* may protect upper surfaces of the vertical active patterns **109***m***1**, **109***p***1**, **109***p***2**, and **109***m***2** during the process of patterning the semiconductor layer (**109** in FIG. **16**A). Openings **130** may be formed by etching the semiconductor layer (**109** in FIG. **16**A). The openings **130** may expose the buffer insulating layer **106**.
[0305] Referring to FIG. **16**C, a gate dielectric layer **136** conformally covering internal walls of the openings **130** may be formed, a preliminary gate conductive layer conformally covering the gate dielectric layer **136** may be formed, gate conductive layers **138** may be formed by anisotropically etching the preliminary gate conductive layer, and gate isolation insulating layers **142** filling the openings **130** may be formed on the gate conductive layers **138**. The mask pattern **127** may be removed.
[0306] Referring to FIG. **16**D, gate electrodes **139***m***1**, **139***p***1**, **139***p***2**, and **139***m***2** may be formed by partially etching the gate conductive layers **138**. The gate electrodes **139***m***1**, **139***p***1**, **139***p***2**, and

139*m***2** may include first word lines **139***m***1** in the first memory region MR**1**, second word lines **139***m***2** in the second memory region MR**2**, and peripheral NMOS gate electrodes **139***p***1** and peripheral PMOS gate electrodes **139***p***2** in the first peripheral circuit region PR_U. Upper gate capping insulating layers **145** may be formed on the gate electrodes **139***m***1**, **139***p***1**, **139***p***2**, and **139***m***2**.

[0307] Referring to FIG. **16**E, a planarization process may be performed until the upper surfaces of the vertical active patterns **109***m***1**, **109***p***1**, **109***p***2**, and **109***m***2** are exposed. Accordingly, the insulating patterns **112***a* may be removed.

[0308] By performing the first ion implantation process, upper cell source/drain regions **187***m* may be formed in upper regions of the first and second cell vertical active patterns **109***m***1** and **109***m***2**. The upper cell source/drain regions **187***m* may have N-type conductivity.

[0309] By performing the second ion implantation process, upper NMOS source/drain regions **187***p***1** may be formed in the upper regions of the NMOS vertical active patterns **109***p***1**. The upper NMOS source/drain regions **187***p***1** may have N-type conductivity.

[0310] In an example, the upper NMOS source/drain regions **187***p***1** may have a doping concentration different from that of the upper cell source/drain regions **187***m*.

[0311] In another example, when the upper NMOS source/drain regions **187***p***1** has the same doping concentration as that of the upper cell source/drain regions **187***m*, the first and second ion implantation processes may be performed as an integrated process.

[0312] By performing the third ion implantation process, upper PMOS source/drain regions **187***p***2** may be formed in the upper regions of the PMOS vertical active patterns **109***p***2**. The upper PMOS source/drain regions **187***p***2** may have P-type conductivity.

[0313] Referring to FIG. **16**F, the insulating structure **151** may be formed on the element formed as above, and plugs **154***m***1**, **154***p***1**, **154***p***2**, and **154***m***2** penetrating the insulating structure **151** may be formed.

[0314] The plugs **154***m***1**, **154***p***1**, **154***p***2**, and **154***m***2** may include cell plug patterns **154***m***1**, **154***m***2** connected to the first and second cell upper source/drain regions **187***m*, and peripheral contact structures **154***p***1** and **154***p***2** connected to the upper NMOS source/drain regions **187***p***1** and the upper PMOS source/drain regions **187***p***2**.

[0315] The insulating structure **157** and the conductive patterns **160***m***1**, **160***pa*, **160***pb*, and **160***m***2** may be formed on the insulating structure **151** and the plugs **154***m***1**, **154***p***1**, **154***p***2**, and **154***m***2**. The insulating structure **157** may be formed on side surfaces of the conductive patterns **160***m***1**, **160***pa*, **160***pb*, and **160***m***2**.

[0316] The conductive patterns **160***m***1**, **160***pa*, **160***pb*, and **160***m***2** may include the cell pad patterns **160***m***1**, **160***m***2** connected to the cell plug patterns **154***m***1**, **154***m***2**, and the upper peripheral wirings **160***pa* and **160***pb* connected to the peripheral contact structures **154***p***1** and **154***p***2**.

[0317] Data storage structures DS**1** and DS**2** may be formed on the cell plug patterns **154***m***1** and **154***m***2**. The data storage structures DS**1** and DS**2** may have first electrodes **163** extending in the vertical direction Z, a second electrode **169** on a side surface and and upper surface of each of the first electrodes **163**, and a dielectric layer between the first electrodes **163** and the second electrode **169**.

[0318] An insulating structure **175** covering the data storage structures DS**1** and DS**2** may be formed, and contact plugs **178***m* and **178***p* penetrating the insulating structure **175** may be formed.

[0319] The contact plugs **178***m*, **178***p* may include cell contact plugs **178***m* connected to the second electrodes **169** of the data storage structures DS**1** and DS**2**, and a peripheral contact plug **178***p* connected to the upper peripheral wiring **160***pb* of one of the upper peripheral wirings **160***pa* and **160***pb*.

[0320] Upper wirings **184***m* and **184***p* connected to the contact plugs **178***m* and **178***p* may be formed on the insulating structure **175**, and upper insulating structure **181** covering the upper wirings **184***m* and **184***p* may be formed.

[0321] Referring to FIG. **16**G, the buffer insulating layer **106**, the vertical active patterns **109***m***1**, **109***p***1**, **109***p***2**, and **109***m***2** and the preliminary back gate electrodes **121** may be exposed by removing the sacrificial substrate **103**.
[0322] By etching back the preliminary back gate electrodes **121**, back gate electrodes **121***m***1**, **121***m***2**, **121***p***1**, and **121***p***2** may be formed.
[0323] The back gate electrodes **121***m***1**, **121***m***2**, **121***p***1**, and **121***p***2** may include cell back gate electrodes **121***m***1** and **121***m***2** in the first and second memory regions MR**1** and MR**2**, and third peripheral gate electrodes **121***p***1** and **121***p***2** in the first peripheral circuit region PR_U.
[0324] Upper back gate capping insulating layers **189** may be formed on the back gate electrodes **121***m***1**, **121***m***2**, **121***p***1**, and **121***p***2**.
[0325] By performing the fourth ion implantation process, lower cell source/drain regions **148***m* may be formed in lower regions of the first and second cell vertical active patterns **109***m***1** and **109***m***2**. The lower cell source/drain regions **148***m* may have N-type conductivity.
[0326] Lower NMOS source/drain regions **148***p***1** may be formed in the lower regions of the NMOS vertical active patterns **109***p***1** by performing the fifth ion implantation process. The lower NMOS source/drain regions **148***p***1** may have N-type conductivity.
[0327] In an example, the lower NMOS source/drain regions **148***p***1** may have a doping concentration different from that of the lower cell source/drain regions **148***m*.
[0328] In another example, when the lower NMOS source/drain regions **148***p***1** has the same doping concentration as the lower cell source/drain regions **148***m*, the first and second ion implantation processes may be performed as an integrated process.
[0329] By performing the sixth ion implantation process, lower PMOS source/drain regions **148***p***2** may be formed in the lower regions of the PMOS vertical active patterns **109***p***2**. The lower PMOS source/drain regions **148***p***2** may have P-type conductivity.
[0330] Each of the cell vertical active patterns **109***m***1** may include the lower cell source/drain region **148***m*, the upper cell source/drain region **187***m*, and the cell channel region CHm between the lower cell source/drain region **148***m* and the upper cell source/drain region **187***m*.
[0331] Each of the NMOS vertical active patterns **109***p***1** may include the lower NMOS source/drain region **148***p***1**, the upper NMOS source/drain region **187***p***1**, and the NMOS channel region CHp**1** between the lower NMOS source/drain region **148***p***1** and the upper NMOS source/drain region **187***p***1**.
[0332] Each of the PMOS vertical active patterns **109***p***2** may include the lower PMOS source/drain region **148***p***2**, the upper PMOS source/drain region **187***p***2**, and the PMOS channel region CHp**2** between the lower PMOS source/drain region **148***p***2** and the upper PMOS source/drain region **187***p***2**.
[0333] Referring to FIG. **16**H, bit lines **190***m***1** and **190***m***2** and lower peripheral wirings **190***p***1** and **190***p***2** may be formed. Each of the bit lines **190***m***1** and **190***m***2** and the lower peripheral wirings **190***p***1** and **190***p***2** may include a first conductive layer **190***a* and a second conductive layer **190***b* on the first conductive layer **190***a*.
[0334] The bit lines **190***m***1** and **190***m***2** may be connected to the cell vertical active patterns **109***m***1**. The lower peripheral wirings **190***p***1** and **190***p***2** may be connected to the peripheral vertical active pattern **109***p***1** and **109***p***2**.
[0335] An insulating structure **198**, a first routing wiring structure **192**, connection contact plugs **194** and first bonding pads **196** may be formed. The first bonding pads **196** may have upper surfaces coplanar with an upper surface of the insulating structure **198**.
[0336] The first routing wiring structure **192** may include a plurality of wirings disposed on different levels and a plurality of vias connected to an upper surface and a lower surface of each of the plurality of wirings. The lower surfaces of the connection contact plugs **194** may be connected to the first upper peripheral wirings **160***pa* among the upper peripheral wirings **160***pa* and **160***pb*. One of the plurality of wirings disposed on different levels of the first routing wiring structure **192**

may be connected to the upper surfaces of the connection contact plugs **194**.
[0337] Accordingly, the first structure CH**1** as in FIGS. **1**A and **7** may be formed.
[0338] In the description below, referring to FIG. **17**, a device isolation region **10***s* limiting active regions **10***a* may be formed on the substrate **5**. The substrate **5** may be configured as a semiconductor substrate. Lower peripheral circuit transistors **25***a*, **25***b*, and **25***c* may be formed on the substrate **5**. Each of the lower peripheral circuit transistors **25***a*, **25***b*, and **25***c* may include peripheral gate structures **15***a* and **15***b* disposed on the active region **10***a*, and peripheral source/drain regions **20***sd* disposed in the active region **10***a* disposed on both sides of the peripheral gate structure **15***a* and **15***b*. The peripheral gate structures **15***a* and **15***b* may include a peripheral gate dielectric layer **15***a* and a peripheral gate electrode **15***b* stacked in order.
[0339] A second routing wiring structure **30** electrically connected to the lower peripheral circuit transistors **25***a*, **25***b*, and **25***c*, second bonding pads **35** electrically connected to the second routing wiring structure **30** on the second routing wiring structure **30**, and a lower insulating structure **40** may be formed on the lower peripheral circuit transistors **25***a*, **25***b*, and **25***c*. The lower insulating structure **40** may have an upper surface coplanar with upper surfaces of the second bonding pads **35**. The lower peripheral circuit transistors **25***a*, **25***b*, and **25***c* may include a first core circuit transistor **25***a* in the first core circuit region CR**1**, a second core circuit transistor **25***b* in the second core circuit region CR**2**, and a second peripheral circuit transistor **25***c* in the second peripheral circuit region PR_L. Accordingly, the second structure CH**2** as in FIG. **1**A may be formed.
[0340] Referring to FIG. **1**A along with FIGS. **16**H and **17**, the first structure CH**1** and the second structure CH**2** may be bonded to each other by performing a wafer bonding process. Upper surfaces of the second bonding pads **35** may be bonded to lower surfaces of the first bonding pads **196**, and upper surfaces of the lower insulating structure **40** may be bonded to lower surfaces of the first insulating structure **198**. Accordingly, a bonding surface JSa may be formed between the first structure CH**1** and the second structure CH**2**.
[0341] According to the aforementioned example embodiments, transistors including vertical active patterns extending in the vertical direction and gate electrodes facing the side surfaces of the vertical active patterns may be provided. Accordingly, integration density of a semiconductor device may increase.
[0342] Also, to increase integration density of the semiconductor device, a first peripheral circuit region may be provided on an outside of the memory region, and a core circuit region may be provided in a region vertically overlapping the memory region.
[0343] Also, to increase integration density of the semiconductor device, a second peripheral circuit region vertically overlapping the first peripheral circuit region may be provided.
[0344] While the example embodiments have been illustrated and described above, it will be configured as apparent to those skilled in the art that modifications and variations may be made without departing from the scope in the example embodiment as defined by the appended claims.

Claims

1. A semiconductor device, comprising: a first structure including a memory region and a first peripheral circuit region; and a second structure vertically overlapping the first structure and including a core circuit region and a second peripheral circuit region, wherein the memory region includes: a first cell vertical active pattern; a first word line having a side surface facing a first side surface of the first cell vertical active pattern; and a first cell gate dielectric layer between the first cell vertical active pattern and the first word line, wherein the first peripheral circuit region includes: a first peripheral vertical active pattern disposed at substantially the same level as the first cell vertical active pattern; a first peripheral gate electrode having a side surface facing a first side surface of the first peripheral vertical active pattern; and a first peripheral gate dielectric layer between the first peripheral vertical active pattern and the first peripheral gate electrode.

2. The semiconductor device of claim 1, wherein the memory region further includes a data storage structure, wherein the core circuit region includes a core circuit transistor vertically overlapping the memory region, and wherein the second peripheral circuit region includes a peripheral circuit transistor not vertically overlapping the memory region.
3. The semiconductor device of claim 2, wherein the first peripheral circuit region further includes a power capacitor disposed at substantially the same level as the data storage structure.
4. The semiconductor device of claim 1, wherein the first cell vertical active pattern includes: a lower cell source/drain region; an upper cell source/drain region on the lower cell source/drain region; and a cell channel region between the lower cell source/drain region and the upper cell source/drain region, wherein the first peripheral vertical active pattern includes: a lower peripheral source/drain region; an upper peripheral source/drain region on the lower peripheral source/drain region; and a peripheral channel region between the lower peripheral source/drain region and the upper peripheral source/drain region, wherein the side surface of the first word line faces the cell channel region of the first cell vertical active pattern, and wherein the side surface of the first peripheral gate electrode faces the peripheral channel region of the first peripheral vertical active pattern.
5. The semiconductor device of claim 1, wherein the memory region further includes: a bit line below the first cell vertical active pattern; a cell contact structure on the first cell vertical active pattern; and a data storage structure on the cell contact structure, wherein the first peripheral circuit region further includes: a first peripheral wiring below the first peripheral vertical active pattern; a peripheral contact structure on the first peripheral vertical active pattern; and a second peripheral wiring on the peripheral contact structure, and wherein the first peripheral wiring is disposed at substantially the same level as the bit line.
6. The semiconductor device of claim 5, wherein the cell contact structure includes: a cell plug pattern on the first cell vertical active pattern; and a cell pad pattern on the cell plug pattern, wherein the peripheral contact structure is disposed at substantially the same level as the cell plug pattern, and wherein the second peripheral wiring is disposed at substantially the same level as the cell pad pattern.
7. The semiconductor device of claim 5, wherein the data storage structure includes: a first electrode extending in a vertical direction and on the cell contact structure; a second electrode on the first electrode; and a dielectric layer between the first electrode and the second electrode, and wherein at least a portion of the second peripheral wiring is disposed at the same level as a portion of the second electrode.
8. The semiconductor device of claim 1, wherein a vertical length of the first peripheral gate electrode is substantially the same as a vertical length of the first word line.
9. The semiconductor device of claim 1, wherein the first peripheral circuit region further includes: a second peripheral vertical active pattern disposed at substantially the same level as the first cell vertical active pattern; a second peripheral gate electrode having a side surface facing a first side surface of the second peripheral vertical active pattern; and a second peripheral gate dielectric layer between the second peripheral vertical active pattern and the second peripheral gate electrode, and wherein a vertical length of the second peripheral gate electrode is different from a vertical length of the first word line.
10. The semiconductor device of claim 1, wherein a thickness of the first peripheral gate dielectric layer is different from a thickness of the first cell gate dielectric layer.
11. The semiconductor device of claim 1, wherein the memory region further includes: a second cell vertical active pattern adjacent to the first cell vertical active pattern and spaced apart from each other; a second word line having a side surface facing a second side surface of the second cell vertical active pattern; a second cell gate dielectric layer between the second cell vertical active pattern and the second word line; and a cell back gate electrode disposed between the first cell vertical active pattern and the second cell vertical active pattern, and wherein the first and second

cell vertical active patterns are disposed between the first word line and the second word line.

12. The semiconductor device of claim 1, wherein the first peripheral circuit region further includes: a second peripheral vertical active pattern adjacent to the first peripheral vertical active pattern and spaced apart from each other; a second peripheral gate electrode having a side surface facing a second side surface of the second peripheral vertical active pattern; a second peripheral gate dielectric layer between the second peripheral vertical active pattern and the second peripheral gate electrode; and a peripheral back gate electrode disposed between the first peripheral vertical active pattern and the second peripheral vertical active pattern, and wherein the first and second peripheral vertical active patterns are disposed between the first peripheral gate electrode and the second peripheral gate electrode.

13. A semiconductor device, comprising: a first structure including a first memory region and a second memory region spaced apart from each other, and a first peripheral circuit region between the first memory region and the second memory region; and a second structure vertically overlapping and bonded to the first structure, wherein the second structure includes: a first core circuit region vertically overlapping the first memory region; a second core circuit region vertically overlapping the second memory region; and a second peripheral circuit region vertically overlapping the first peripheral circuit region, wherein each of the first and second memory regions includes a cell transistor and a data storage structure electrically connected to the cell transistor, wherein the cell transistor includes: a cell vertical active pattern; a word line having a side surface facing at least a portion of a side surface of the cell vertical active pattern; and a cell gate dielectric layer between the cell vertical active pattern and the word line.

14. The semiconductor device of claim 13, wherein the first peripheral circuit region includes: a peripheral vertical active pattern disposed at substantially the same level as the cell vertical active pattern; a peripheral gate electrode having a side surface facing at least a portion of a side surface of the peripheral vertical active pattern; and a peripheral gate dielectric layer between the peripheral vertical active pattern and the peripheral gate electrode.

15. The semiconductor device of claim 13, wherein the first peripheral circuit region further includes a power capacitor disposed at substantially the same level as the data storage structure.

16. The semiconductor device of claim 13, wherein the first core circuit region includes a first core circuit transistor vertically overlapping the data storage structure of the first memory region, wherein the second core circuit region includes a second core circuit transistor vertically overlapping the data storage structure of the second memory region, and wherein the second peripheral circuit region includes a peripheral circuit transistor not vertically overlapping the data storage structures of the first and second memory regions.

17. A semiconductor device, comprising: a memory region including a cell transistor and data storage structure; a first peripheral circuit region disposed on an outside of the memory region and including a first peripheral circuit transistor; a core circuit region vertically overlapping the memory region and including a core circuit transistor; and a second peripheral circuit region disposed on an outside of the core circuit region, vertically overlapping the first peripheral circuit region, and including a second peripheral circuit transistor.

18. The semiconductor device of claim 17, wherein the cell transistor includes: a cell vertical active pattern; and a word line having a side surface facing a first side surface of the cell vertical active pattern, and wherein the first peripheral circuit transistor includes: a peripheral vertical active pattern disposed at substantially the same level as the cell vertical active pattern; and a peripheral gate electrode having a side surface facing a first side surface of the peripheral vertical active pattern.

19. The semiconductor device of claim 18, wherein the memory region further includes a bit line disposed below the cell vertical active pattern, and wherein the first peripheral circuit region further includes a peripheral wiring disposed below the peripheral vertical active pattern and disposed at substantially the same level as the bit line.

20. The semiconductor device of claim 17, wherein the first peripheral circuit region further includes a power capacitor disposed at substantially the same level as the data storage structure.